ALGEBRA I · COMMON CORE HIGH SCHOOL MATH

Question Collection

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

84 question sets • 2100 questions

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SET 1 • HSN-RN.A.1 • NUMBER & QUANTITY

Explain the meaning of rational exponents from the properties of integer exponents

MAIN	Algebra I main question	**HSN-RN.A.1**
BACKWARD 1	Apply the properties of integer exponents	**8.EE.A.1**
BACKWARD 2	Evaluate square roots and cube roots	**8.EE.A.2**
FORWARD 1	Express the solution of an exponential equation as a logarithm	**HSF-LE.A.4**
FORWARD 2	Use the inverse relationship between exponents and logarithms	**HSF-BF.B.5**

SET 1 • HSN-RN.A.1

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

For the power-of-a-power property to hold, $(7^{1/4})^4$ must equal $7^{(1/4)\cdot 4} = 7$.

What number must $7^{1/4}$ be? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q1/5

Use the properties of exponents to explain why $8^{2/3} = 4$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q2/5

For $x > 0$, which expression is equal to $x^{3/4}$?

A. $x^3 \div 4$

B. the fourth root of x, cubed

C. the cube root of x, to the fourth power

D. $3x \div 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q3/5

True or false?

$16^{1/2} = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q4/5

True or false?

For $x > 0$, $x^{1/2} \cdot x^{1/2} = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSN-RN.A.1

BACKWARD 1

Apply the properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$2^3 \cdot 2^5$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q1/5

Write as a single power.

$(3^2)^4$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q2/5

Write as a single power.

$5^6 \div 5^2$

A. 1^4

B. 5^4

C. 5^3

D. 5^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q3/5

True or false?

$(7^2)^3 = 7^6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q4/5

True or false?

$4^0 = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • HSN-RN.A.1

BACKWARD 2

Evaluate square roots and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Find the value.

$\sqrt{81}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

Find the value.

$\sqrt[3]{27}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

Find the value.

$\sqrt[3]{64}$

A. 32

B. 4

C. 8

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

True or false?

$\sqrt[3]{(-8)} = -2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

True or false?

$\sqrt{25} = \pm 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSN-RN.A.1

FORWARD 1

Express the solution of an exponential equation as a logarithm • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $10^x = 500$. Write the solution as a logarithm, then use a calculator to round it to the nearest hundredth.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q1/5

A colony of 400 bacteria grows by a factor of 10 every hour: $N = 400 \cdot 10^{t}$. Solve $400 \cdot 10^{t} = 3{,}000$. Give the exact logarithm and round to the nearest hundredth. (log means base 10.)

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q2/5

Which expression is the solution of $3 \cdot 10^x = 60$? (log means base 10.)

A. $x = \log 60$

B. $x = 20$

C. $x = \log 57$

D. $x = \log 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q3/5

True or false?

The solution of $2 \cdot 10^t = 1{,}000$ is $t = \log 500 \approx 2.70$. (log means base 10.)

A. True

B. False

True or false?

The solution of $10^x = -10$ is $x = -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q5**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSN-RN.A.1

FORWARD 2

Use the inverse relationship between exponents and logarithms • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

$f(x) = 10^x$ and $g(x) = \log x$ (log means base 10).

Find $f(g(100))$ and $g(f(2))$, and say what they show.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q1/5

A sound level is $L = 10 \log(I \div I_0)$ decibels, where log means base 10.

Solve for the intensity I in terms of L.

$I =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q2/5

Which function is the inverse of $f(x) = 5^x$?

A. $f^{-1}(x) = x \div 5$

B. $f^{-1}(x) = x^5$

C. $f^{-1}(x) = \log_5 x$

D. $f^{-1}(x) = 5^{-x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q3/5

True or false?

$10^{\log_{10} 7} = 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q4/5

True or false?

$\log(10^x) = 10x$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q5/5

SET 2 • HSN-RN.A.2 • NUMBER & QUANTITY

Rewrite expressions with radicals and rational exponents

MAIN	Algebra I main question	**HSN-RN.A.2**
BACKWARD 1	Solve equations of the form $x^2 = p$ and $x^3 = p$	**8.EE.A.2**
BACKWARD 2	Use the meaning of rational exponents	**HSN-RN.A.1**
FORWARD 1	Solve radical equations and identify extraneous solutions	**HSA-REI.A.2**
FORWARD 2	Add, subtract, and multiply complex numbers	**HSN-CN.A.2**

SET 2 • HSN-RN.A.2

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

Assume $x \geq 0$. Write $x^{5/2}$ using a radical.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q1/5

Write $\sqrt[3]{(y^2)}$ using a rational exponent.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q2/5

Simplify.

$\sqrt{50} \cdot \sqrt{2}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q3/5

Which expression is equal to $(27x^6)^{1/3}$ for every real number x?

A. $27x^2$

B. $9x^2$

C. $3x^3$

D. $3x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q4/5

True or false?

$\sqrt{(x^2)} = x$ for every real number x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q5/5

SET 2 • HSN-RN.A.2

BACKWARD 1

Solve equations of the form $x^2 = p$ and $x^3 = p$ • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve for x.

$x^3 = 125$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q1/5

Solve for x. Give all solutions.

$x^2 = 36$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q2/5

Solve for x.

$x^3 = -8$

A. ± 2

B. -2

C. -4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q3/5

True or false?

$x^2 = 7$ has the solutions $\sqrt{7}$ and $-\sqrt{7}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q4/5

True or false?

$x^3 = 27$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSN-RN.A.2

BACKWARD 2

Use the meaning of rational exponents • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

Find the value.

$25^{1/2}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q1/5

Find the value.

$27^{2/3}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q2/5

Find the value.

$32^{1/5}$

A. 160

B. 2

C. 6.4

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-B2-Q3**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$9^{3/2} = 27$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q4/5

True or false?

$8^{1/3} = \frac{8}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSN-RN.A.2

FORWARD 1

Solve radical equations and identify extraneous solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $\sqrt{(x + 3)} = 5$. Check your solution.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q1/5

Solve $\sqrt{(2x + 1)} = x - 1$. Check for extraneous solutions.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q2/5

What are the solutions of $\sqrt{x} = -3$?

A. $x = 3$

B. $x = 9$

C. $x = -9$

D. There is no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q3/5

True or false?

Squaring both sides of an equation can introduce extraneous solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q4**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q4/5

True or false?

$x = -4$ is a solution of $\sqrt{(x + 8)} = x + 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q5**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSN-RN.A.2

FORWARD 2

Add, subtract, and multiply complex numbers • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2

Add. Write the answer in the form a + bi.

$(3 + 2i) + (1 - 5i)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q1/5

Multiply. Write the answer in the form a + bi.

$(2 + i)(3 - i)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q2/5

Simplify.

$i^2 \cdot i$

A. 1

B. $-i$

C. i

D. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q3**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q3/5

True or false?

$(1 + i)^2 = 2i$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q4/5

True or false?

$i \cdot i = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q5**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q5/5

SET 3 • HSN-RN.B.3 • NUMBER & QUANTITY

Prove that the sum of a rational number and an irrational number is irrational

MAIN	Algebra I main question	**HSN-RN.B.3**
BACKWARD 1	Classify numbers as rational or irrational	**8.NS.A.1**
BACKWARD 2	Write rational numbers as quotients of integers	**7.NS.A.2.b**
FORWARD 1	Find exact lengths and perimeters on the coordinate plane	**HSG-GPE.B.7**
FORWARD 2	Write numbers in the complex number system as a + bi	**HSN-CN.A.1**

SET 3 • HSN-RN.B.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3

Explain why $5 + \sqrt{2}$ is irrational. (Suppose $5 + \sqrt{2} = r$ for some rational number r.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q1/5

Explain why $\sqrt{3} - 4$ is irrational. (Suppose $\sqrt{3} - 4 = r$ for some rational number r.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q2/5

Prove that if a is rational and x is irrational, then a + x is irrational.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q3/5

Which sum is irrational?

A. $\sqrt{2} + (-\sqrt{2})$

B. $0.25 + \frac{3}{4}$

C. $\sqrt{16} + 1.5$

D. $\sqrt{5} + 1.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q4/5

True or false?

The sum of two irrational numbers is always irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q5/5

SET 3 • HSN-RN.B.3

BACKWARD 1

Classify numbers as rational or irrational • Grade 8 • The Number System • 8.NS.A.1

Is $\sqrt{49}$ rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q1/5

The number 0.1010010001... continues with one more 0 before each 1. Is it rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q2/5

Which number is rational?

A. $\sqrt{2}$

B. $\sqrt{10}$

C. π

D. 0.2727... (the digits 27 repeat)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

Every integer is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q4/5

True or false?

$\sqrt{20}$ is rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSN-RN.B.3

BACKWARD 2

Write rational numbers as quotients of integers • Grade 7 • The Number System • 7.NS.A.2.b

Write 1.25 as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q1/5

Write $-3\frac{1}{3}$ as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q2/5

Which quotient is equal to 0.6?

A. $\frac{6}{100}$

B. $\frac{5}{3}$

C. $\frac{3}{5}$

D. $\frac{1}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q3/5

True or false?

$$-7 = \frac{-7}{1}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q4/5

True or false?

$\frac{2}{0}$ is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSN-RN.B.3

FORWARD 1

Find exact lengths and perimeters on the coordinate plane • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the exact perimeter of the triangle with vertices (0, 0), (3, 0), and (0, 3).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q1/5

A square has vertices (0, 0), (1, 1), (0, 2), and (−1, 1). Find its exact perimeter, and say whether it is rational or irrational.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q2/5

What is the area of the triangle with vertices (0, 0), (4, 0), and (4, 3)?

A. 6

B. 12

C. 7

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q3/5

True or false?

The triangle with vertices (0, 0), (1, 0), and (0, 1) has an irrational perimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q4/5

True or false?

The triangle with vertices (0, 0), (6, 0), and (0, 8) has an irrational perimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSN-RN.B.3

FORWARD 2

Write numbers in the complex number system as a + bi • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{-49}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q1/5

Name the real part and the imaginary part of $4 - 3i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q2/5

Which of these is a real number?

A. i

B. i^2

C. $2i$

D. $1 + i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q3**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q3/5

True or false?

Every real number is a complex number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q4/5

True or false?

$i^2 = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q5/5

SET 69 • HSN-RN.B.3 • NUMBER & QUANTITY

Prove that the product of a nonzero rational number and an irrational number is irrational

MAIN	Algebra I main question	**HSN-RN.B.3**
BACKWARD 1	Classify numbers as rational or irrational	**8.NS.A.1**
BACKWARD 2	Multiply and divide rational numbers	**7.NS.A.2.c**
FORWARD 1	Find exact lengths, perimeters, and areas with irrational side lengths	**HSG-GPE.B.7**
FORWARD 2	Factor sums of squares over the complex numbers	**HSN-CN.C.8**

SET 69 • HSN-RN.B.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3

Explain why $3\sqrt{7}$ is irrational. (Suppose $3\sqrt{7} = r$ for some rational number r.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q1/5

Prove that if a is a nonzero rational number and x is irrational, then ax is irrational.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q2/5

Why does the statement "a rational number times an irrational number is irrational" need the rational number to be nonzero?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q3/5

Which product is irrational?

A. $\sqrt{5} \cdot \sqrt{5}$

B. $\sqrt{2} \cdot \sqrt{8}$

C. $0 \cdot \sqrt{5}$

D. $\frac{2}{3}\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q4/5

True or false?

The product of two irrational numbers is always irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q5/5

SET 69 • HSN-RN.B.3

BACKWARD 1

Classify numbers as rational or irrational • Grade 8 • The Number System • 8.NS.A.1

Is √64 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q1/5

The decimal 0.1010010001... has one more 0 between the 1s each time. Is it rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q2/5

Which of these numbers is irrational?

A. $\sqrt{10}$

B. $-\frac{4}{7}$

C. $\sqrt{100}$

D. 0.333...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q3/5

True or false?

Every terminating decimal is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q4/5

True or false?

$\pi = \frac{22}{7}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • HSN-RN.B.3

BACKWARD 2

Multiply and divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find $(-\frac{2}{3}) \cdot \frac{9}{4}$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q1/5

Find $\frac{5}{6} \div \left(-\frac{10}{3}\right)$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q2/5

Which product is positive?

A. $(0.5)(-8)$

B. $0 \cdot 8$

C. $(-0.5)(-8)$

D. $(-0.5)(8)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q3/5

True or false?

The quotient of two nonzero rational numbers is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q4/5

True or false?

$\frac{3}{4} \div 0 = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q5/5

SET 69 • HSN-RN.B.3

FORWARD 1

Find exact lengths, perimeters, and areas with irrational side lengths • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

A square has vertices (0, 0), (2, 1), (1, 3), and (−1, 2). Find its exact side length and its area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q1/5

A rectangle has vertices (0, 0), (3, 3), (1, 5), and (−2, 2). Find its exact perimeter.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q2/5

What is the exact perimeter of the triangle with vertices (0, 0), (4, 0), and (0, 4)?

A. $4\sqrt{2}$

B. 16

C. $8 + 2\sqrt{2}$

D. $8 + 4\sqrt{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q3/5

True or false?

A square with side $\sqrt{5}$ has a rational area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q4/5

True or false?

A square with side $\sqrt{5}$ has a rational perimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q5/5

SET 69 • HSN-RN.B.3

FORWARD 2

Factor sums of squares over the complex numbers • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^2 + 9$ over the complex numbers.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q1/5

Expand $(x + 2i)(x - 2i)$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q2/5

Which is a factorization of $x^2 + 25$?

A. $(x + 5i)(x - 5i)$

B. $(x + 5)(x - 5)$

C. $(x + 5i)^2$

D. $(x - 5i)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q3/5

True or false?

$$x^2 + 16 = (x + 4i)(x - 4i)$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F2-Q4**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q4/5

True or false?

$x^2 + 4$ cannot be factored, even with complex numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q5/5

SET 70 • HSN-RN.B.3 • NUMBER & QUANTITY

Prove that sums and products of two rational numbers are rational

MAIN	Algebra I main question	**HSN-RN.B.3**
BACKWARD 1	Write quotients of integers as rational numbers	**7.NS.A.2.b**
BACKWARD 2	Divide fractions	**6.NS.A.1**
FORWARD 1	Show that sums and products of complex numbers are complex n…	**HSN-CN.A.2**
FORWARD 2	Show that rational expressions are closed under addition and mult…	**HSA-APR.D.7**

SET 70 • HSN-RN.B.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3

Explain why the sum of two rational numbers a/b and c/d (with integers a, b, c, d, and b ≠ 0, d ≠ 0) is rational.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q1/5

Prove that the product of two rational numbers a/b and c/d (with integers a, b, c, d, and $b \neq 0$, $d \neq 0$) is rational.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q2/5

In the proof that (a/b)(c/d) is rational, why is bd ≠ 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q3/5

Which statement shows that $(\frac{2}{5})(-\frac{15}{4})$ is rational?

A. It can be written as a decimal with a calculator.

B. It equals $-30/20 = -3/2$, a quotient of integers with a nonzero denominator.

C. It is negative.

D. Its first factor has an even numerator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q4/5

True or false?

The difference of two rational numbers is always rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q5/5

SET 70 • HSN-RN.B.3

BACKWARD 1

Write quotients of integers as rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Write $-\frac{3}{4}$ as a quotient of integers in two different ways.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q1/5

Is $-12 \div 5$ a rational number? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q2/5

Which quotient is undefined?

A. $-7 \div 2$

B. $0 \div 7$

C. $7 \div 0$

D. $7 \div (-1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q3/5

True or false?

For integers p and q with $q \neq 0$, $-(p/q) = (-p)/q = p/(-q)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q4/5

True or false?

Some integers are not rational numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q5/5

SET 70 • HSN-RN.B.3

BACKWARD 2

Divide fractions • Grade 6 • The Number System • 6.NS.A.1

Find $\frac{3}{4} \div \frac{1}{8}$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q1/5

Find $\frac{2}{3} \div \frac{5}{6}$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q2/5

Which number equals $\frac{5}{2} \div \frac{1}{4}$?

A. $\frac{5}{8}$

B. $\frac{1}{10}$

C. 10

D. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{2} \div \frac{1}{4} = 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q4/5

True or false?

$\frac{3}{5} \div 3 = \frac{9}{5}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • HSN-RN.B.3

FORWARD 1

Show that sums and products of complex numbers are complex numbers • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2

Add. Write the answer in the form a + bi.

$(3 - 2i) + (-1 + 6i)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q1/5

Multiply. Write the answer in the form a + bi.

$(2 + i)(3 + 2i)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q2/5

Which shows that the product of two complex numbers is a complex number?

A. $i^2 = 1$

B. $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$, and $ac - bd$ and $ad + bc$ are real numbers

C. $(a + bi)(c + di) = ac + bdi$

D. The product of two complex numbers is always real

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q3/5

True or false?

The sum of two complex numbers is a complex number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q4/5

True or false?

$(1 + i) + (1 - i)$ is not a complex number because it has no i.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q5/5

SET 70 • HSN-RN.B.3

FORWARD 2

Show that rational expressions are closed under addition and multiplication • High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Assume $x \neq 0$ and $x \neq -1$. Write $1 \div x + 2 \div (x + 1)$ as one fraction.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q1/5

Assume $x \neq 2$ and $x \neq -3$. Multiply and simplify:
$(x \div (x - 2)) \cdot ((x - 2) \div (x + 3))$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q2/5

Why is the sum of two rational expressions a rational expression?

A. Over a common denominator, the sum is a polynomial divided by a nonzero polynomial.

B. The denominators always cancel.

C. Rational expressions are always integers.

D. Any sum of fractions equals 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q3/5

True or false?

The quotient of two rational expressions, when the divisor is not 0, is a rational expression.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q4/5

True or false?

$1 \div x + 1 \div y = 2 \div (x + y)$ for all nonzero x and y with $x + y \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q5/5

SET 4 • HSN-Q.A.1 • NUMBER & QUANTITY

Use units to guide multistep solutions and to choose scales for graphs

MAIN	Algebra I main question	**HSN-Q.A.1**
BACKWARD 1	Convert measurement units with ratio reasoning	**6.RP.A.3.d**
BACKWARD 2	Compute unit rates with fractions	**7.RP.A.1**
FORWARD 1	Use density in a model	**HSG-MG.A.2**
FORWARD 2	Solve problems with velocity vectors — nearest related	**HSN-VM.A.3**

SET 4 • HSN-Q.A.1

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1

A car goes 32 miles on each gallon of gas, and gas costs $3.60 per gallon.

Use units to find the cost of gas for a 400-mile trip. Show how the units cancel.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q1/5

Convert 72 kilometers per hour to meters per second.

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q2/5

The formula d = rt gives distance. A rate r is in meters per second and a time t is in minutes.

What must be done before multiplying?

A. Divide r by t.

B. Convert t to seconds (or r to meters per minute).

C. Nothing; the units already match.

D. Convert d to minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q3/5

A graph will show a town's population each year from 1990 to 2020. The populations range from 12,000 to 31,000.

Choose a scale for the vertical axis, and explain your choice.

Scale: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q4/5

True or false?

A speed of 5 miles per hour is about 7.3 feet per second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q5/5

SET 4 • HSN-Q.A.1

BACKWARD 1

Convert measurement units with ratio reasoning • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

How many inches are in 3.5 feet?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q1/5

How many kilograms are in 2,500 grams?

Kilograms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q2/5

How many feet are in 4 yards?

A. 48

B. 36

C. $\frac{4}{3}$

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q3/5

True or false?

1 hour = 3,600 seconds

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q4/5

True or false?

5 km = 500 m

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSN-Q.A.1

BACKWARD 2

Compute unit rates with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A walker goes $\frac{3}{4}$ mile in $\frac{1}{4}$ hour.

How many miles per hour is this?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q1/5

Half a pound of cheese costs $3.

What is the price per pound?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q2/5

$\frac{2}{3}$ gallon of paint covers $\frac{1}{3}$ of a fence.

How many gallons cover the whole fence?

A. $\frac{1}{2}$

B. $\frac{2}{9}$

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q3/5

True or false?

Walking $\frac{1}{2}$ mile in $\frac{1}{6}$ hour is a speed of 3 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q4/5

True or false?

Painting $\frac{1}{4}$ of a wall in $\frac{1}{2}$ hour is a rate of $\frac{1}{8}$ wall per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSN-Q.A.1

FORWARD 1

Use density in a model • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2

Copper has a density of 8.96 grams per cubic centimeter.

What is the mass of a 5 cubic-centimeter piece of copper?

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q1/5

A county has 240,000 people living in 600 square miles.

What is its population density?

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q2/5

A 2 cm by 3 cm by 4 cm block has a mass of 60 grams.

What is its density?

A. 0.4 gram per cubic centimeter

B. 60 grams

C. 2.5 grams per cubic centimeter

D. 24 grams per cubic centimeter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q3/5

True or false?

Population density is the number of people divided by the area they live on.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q4/5

True or false?

An object with a volume of 10 cubic centimeters and a density of 3 grams per cubic centimeter has a mass of 3.3 grams.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSN-Q.A.1

FORWARD 2

Solve problems with velocity vectors • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Precalculus standard. Connection: Velocity vectors are rate quantities with units, like the unit rates of the MAIN set, with a direction added.

A boat heads east at 12 km/h across a river whose current flows north at 5 km/h.

What is the boat's resulting speed?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q1/5 • NEAREST RELATED

A plane points north at 300 km/h, and the wind blows west at 40 km/h.

Write its velocity as (east part, north part) in km/h.

Velocity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q2/5 • NEAREST RELATED

Two velocities, (3, 4) and (5, −1), are added.

What is the result?

A. (8, 5)

B. (8, 3)

C. (15, −4)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

An object with velocity (6, 8) meters per second has a speed of 10 meters per second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

Adding the velocities (3, 0) and (0, 4) gives a speed of 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 5 • HSN-Q.A.2 • NUMBER & QUANTITY

Define appropriate quantities for descriptive modeling

MAIN	Algebra I main question	**HSN-Q.A.2**
BACKWARD 1	Use variables to represent quantities in a situation	**6.EE.B.6**
BACKWARD 2	Write and solve px + q = r from a word problem	**7.EE.B.4.a**
FORWARD 1	Use geometric quantities to make design decisions	**HSG-MG.A.3**
FORWARD 2	Recognize surveys, experiments, and observational studies and th…	**HSS-IC.B.3**

SET 5 • HSN-Q.A.2

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2

A school wants to model the total cost of a field trip by bus.

Name the quantities the model needs, with a unit for each.

Quantities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q1/5

A club rents a room for $300 and pays $40 for each member who attends. Its cost model is C = 40n + 300.

Define C and n, with units.

Definitions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q2/5

Which quantity best describes how fast a phone battery runs down?

A. Number of apps on the phone

B. Percent of charge used per hour

C. Width of the screen in inches

D. Mass of the phone in grams

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q3/5

Two towns have different populations and different areas.

Define one quantity, with units, that compares how crowded they are.

Quantity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q4/5

True or false?

Kilowatt-hours per month is a sensible quantity for a model of a household's electricity use.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q5/5

SET 5 • HSN-Q.A.2

BACKWARD 1

Use variables to represent quantities in a situation • Grade 6 • Expressions & Equations • 6.EE.B.6

Tickets cost $8 each.

Write an expression for the cost of n tickets.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q1/5

A worker earns $15 per hour plus a $20 bonus.

Write an expression for the pay for h hours.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q2/5

A typist types 30 words per minute.

Which expression gives the number of words typed in m minutes?

A. $\frac{m}{30}$

B. $m + 30$

C. $30m$

D. $\frac{30}{m}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q3/5

True or false?

If w is a number of weeks, 7w is the number of days.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q4/5

True or false?

If d is a number of dollars, d + 5 is 5 times as many dollars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSN-Q.A.2

BACKWARD 2

Write and solve px + q = r from a word problem • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A gym charges a $25 fee plus $10 per class. Tom paid $95.

How many classes did he take?

Classes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q1/5

A phone costs $50 plus $15 per month. The total so far is $140.

For how many months has it been paid?

Months: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q2/5

A plumber charges a $12 fee plus r dollars per hour. A 3-hour job costs $57.

Which equation fits?

A. $3r + 12 = 57$

B. $3(r + 12) = 57$

C. $12r + 3 = 57$

D. $r + 3 + 12 = 57$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q3/5

True or false?

$x = 6$ is the solution of $4x + 6 = 30$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q4/5

True or false?

$x = 3$ is the solution of $2x - 5 = 11$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSN-Q.A.2

FORWARD 1

Use geometric quantities to make design decisions • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A rectangular garden needs an area of at least 200 square feet and can use at most 60 feet of fence.

Does a 10 ft by 20 ft garden meet both requirements? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q1/5

A box with a square base 10 cm on each side must hold 1,000 cubic centimeters.

How tall must it be?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q2/5

A rectangular pen must have an area of 100 square meters.

Which dimensions use the least fence?

A. 5 m by 20 m

B. 2 m by 50 m

C. 10 m by 10 m

D. 4 m by 25 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q3/5

True or false?

A 6 by 6 square has more area than a 4 by 8 rectangle, even though their perimeters are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q4/5

True or false?

Of all rectangles with a perimeter of 20, the 1 by 9 rectangle has the greatest area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSN-Q.A.2

FORWARD 2

Recognize surveys, experiments, and observational studies and the variables they measure • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Defining the quantities a model needs leads to deciding which variables a study measures and how they are collected.

A researcher randomly assigns 40 plants to two fertilizers and measures each plant's growth in centimeters.

What kind of study is this, and what is the response variable (with units)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.3 • Set 5 • FORWARD 2 • Q1/5 • NEAREST RELATED

A survey asks 200 randomly chosen students how many hours they sleep on a school night.

What kind of study is this, and what variable does it measure?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions

HSS-IC.B.3 • Set 5 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which kind of study can show cause and effect?

A. An observational study

B. A poll of volunteers

C. A randomized experiment

D. A survey

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.3 • Set 5 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

In an observational study, the researchers do not assign treatments.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A survey of volunteers who answer an online poll is a random sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.3 • Set 5 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 6 • HSN-Q.A.3 • NUMBER & QUANTITY

Choose a level of accuracy that fits the limits of measurement

MAIN	Algebra I main question	**HSN-Q.A.3**
BACKWARD 1	Estimate to check whether an answer is reasonable	**7.EE.B.3**
BACKWARD 2	Round decimals to any place	**5.NBT.A.4**
FORWARD 1	Use volume formulas and report results to a sensible accuracy	**HSG-GMD.A.3**
FORWARD 2	Use the margin of error of a sample survey — nearest related	**HSS-IC.B.4**

SET 6 • HSN-Q.A.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3

A ruler measures to the nearest 0.1 cm. A rectangle measures 4.3 cm by 2.7 cm, and a calculator gives its area as 11.61 square cm.

Which report of the area best fits the accuracy of the measurements?

A. 11.610 square cm

B. 11.61 square cm

C. About 10 square cm

D. About 12 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q1/5

A stopwatch reads to the nearest 0.1 second. A runner covers 100 m in 12.4 seconds, and a calculator gives 8.0645161 meters per second.

Report the speed to a sensible accuracy.

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q2/5

A tape measure is marked in centimeters.

Which is the most sensible report of a person's height measured with it?

A. 167.3482 cm

B. 170.00 cm

C. 2 m

D. 167 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q3/5

A bridge is 1,250 m long, measured to the nearest 10 m.

Between which two lengths is its true length?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q4/5

True or false?

Giving more decimal places always makes a measured answer more accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q5/5

SET 6 • HSN-Q.A.3

BACKWARD 1

Estimate to check whether an answer is reasonable • Grade 7 • Expressions & Equations • 7.EE.B.3

Estimate.

4.9×21.2

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q1/5

Is 38.4 ÷ 4.8 = 80 reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q2/5

Which is the best estimate of 19.8% of 503?

A. 100

B. 50

C. 1,000

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q3/5

True or false?

$29.95 × 4 is about $120.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q4/5

True or false?

0.48 × 61 is about 300.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSN-Q.A.3

BACKWARD 2

Round decimals to any place • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 8.0645 to the nearest hundredth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q1/5

Round 11.61 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q2/5

Round 0.0357 to the nearest thousandth.

A. 0.035

B. 0.0357

C. 0.036

D. 0.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q3/5

True or false?

3.149 rounded to the nearest tenth is 3.1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q4/5

True or false?

6.95 rounded to the nearest tenth is 6.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSN-Q.A.3

FORWARD 1

Use volume formulas and report results to a sensible accuracy • High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cylinder has a radius of 3.0 cm and a height of 10.0 cm.

Find its volume to the nearest cubic centimeter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q1/5

A sphere has a radius of 5 m.

Find its volume exactly in terms of π, then to the nearest cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q2/5

A cone has a radius of 6 and a height of 4.

What is its volume?

A. 144π

B. 96π

C. 48π

D. 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q3/5

True or false?

A cylinder with radius 2 cm and height 2 cm has a volume of 8π cm^3, which is about 25 cm^3 to the nearest cubic centimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q4/5

True or false?

Doubling the radius of a sphere doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSN-Q.A.3

FORWARD 2

Use the margin of error of a sample survey • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: The accuracy a measurement supports leads to the accuracy of an estimate from a sample, given by its margin of error.

A poll reports that 54% of voters support a plan, with a margin of error of 3 percentage points.

What interval of values for the true percent does this give?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.4 • Set 6 • FORWARD 2 • Q1/5 • NEAREST RELATED

A poll reports 54% support with a margin of error of 3 percentage points.

Do the results suggest that a majority supports the plan? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q2/5 • NEAREST RELATED

A poll reports 48% support with a margin of error of 4 percentage points.

Which statement fits?

A. The poll must be wrong.

B. Fewer than half support the plan, for certain.

C. The true percent could be above or below 50%.

D. The true percent is exactly 48%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

Larger random samples tend to have smaller margins of error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A margin of error of 3 percentage points means the poll is off by exactly 3 points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q5/5 • NEAREST RELATED

Interpret the terms, factors, and coefficients of an expression in context

MAIN	Algebra I main question	**HSA-SSE.A.1.a**
BACKWARD 1	Identify the parts of an expression	**6.EE.A.2.b**
BACKWARD 2	Rewrite an expression to show how quantities are related	**7.EE.A.2**
FORWARD 1	Rewrite a rational expression and interpret its parts	**HSA-APR.D.6**
FORWARD 2	Compose functions and interpret the composite in context	**HSF-BF.A.1.c**

SET 7 • HSA-SSE.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A gym charges $C = 35 + 20m$ dollars for m months of membership.

What do 35 and 20 represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q1/5

The expression 1.08p gives the total cost, including sales tax, of an item with price p.

What does the factor 1.08 tell you?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q2/5

The perimeter of a rectangle is $2(l + w)$.

What does the factor $(l + w)$ represent?

A. The area

B. The length of one side

C. Half the perimeter: one length plus one width

D. Twice the perimeter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q3/5

For the expression $3x^2 - 5x + 7$, name its terms, the coefficient of x, and the constant term.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q4/5

True or false?

A car's value is $500(0.9)^t$ dollars after t years. The factor 0.9 means the car loses 90% of its value each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 1

Identify the parts of an expression • Grade 6 • Expressions & Equations • 6.EE.A.2.b

How many terms does 4a + 3b − 2 have?

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q1/5

What is the coefficient of y in $7y - 2$?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q2/5

Which description fits 5(x + 2)?

A. A product of two factors, 5 and (x + 2)

B. A sum of three terms

C. A single term with coefficient 2

D. A quotient of two numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q3/5

True or false?

In 8 + 2n, the term 2n has a coefficient of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q4/5

True or false?

The terms of $6(m - 1)$ are 6 and m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 2

Rewrite an expression to show how quantities are related • Grade 7 • Expressions & Equations • 7.EE.A.2

Rewrite a + 0.05a as a single term.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q1/5

A price p is lowered by 20%.

Write the new price as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q2/5

Which expression shows a 15% increase of n?

A. 1.15n

B. 15n

C. n + 15

D. 0.15n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q3/5

True or false?

$p - 0.25p = 0.75p$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q4/5

True or false?

$x + 0.3x = 0.3x^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 1

Rewrite a rational expression and interpret its parts • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

A company's average cost per item for n items is (500 + 3n) ÷ n dollars.

Rewrite it as a sum of two terms, and say what each term means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q1/5

Assume $x \neq 0$. Rewrite $(2x + 5) \div x$ in the form $a + b/x$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q2/5

Assume $x \neq 0$. Which expression is equal to $(x^2 + 3x) \div x$?

A. $x^2 + 3$

B. $x + 3x$

C. $x + 3$

D. $4x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q3/5

True or false?

$(6x + 4) \div 2 = 3x + 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q4/5

True or false?

Assume $x \neq 0$. $(x + 6) \div x = 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 2

Compose functions and interpret the composite in context • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

$T(h) = 20 + 5h$ is a temperature in °C after h hours, and $h(t) = t \div 60$ converts t minutes to hours.

Write $T(h(t))$ in simplest form, and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q1/5

$f(x) = x + 3$ and $g(x) = 2x$.

Find $f(g(4))$.

$f(g(4)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q2/5

$f(x) = x^2$ and $g(x) = x - 1$.

Which expression is $f(g(x))$?

A. $x - 1^2$

B. $x^2 - 1$

C. $x^2 - x$

D. $(x - 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q3/5

True or false?

If f(x) = 3x and g(x) = x + 2, then g(f(1)) = 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q4/5

True or false?

f(g(x)) is always equal to g(f(x)).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q5/5

Interpret complicated expressions by viewing one or more of their parts as a single quantity

MAIN	Algebra I main question	**HSA-SSE.A.1.b**
BACKWARD 1	Apply the distributive property to write equivalent expressions	**6.EE.A.3**
BACKWARD 2	Evaluate expressions for given values	**6.EE.A.2.c**
FORWARD 1	Use the formula for the sum of a finite geometric series	**HSA-SSE.B.4**
FORWARD 2	Solve exponential growth models with logarithms	**HSF-LE.A.4**

SET 8 • HSA-SSE.A.1.b

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

An investment of P dollars is worth $P(1 + r)^n$ dollars after n years, where $r > 0$.

Treating $(1 + r)^n$ as one quantity, explain why the investment grows.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q1/5

For which value of x is $10 - (x - 3)^2$ greatest, and what is that greatest value? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q2/5

Which expression is never greater than 4, for every real number x?

A. $(x + 1)^2 - 4$

B. $4 - (x + 1)^2$

C. $4(x + 1)^2$

D. $4 + (x + 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q3/5

A square picture with side s is surrounded by a frame 2 units wide on every side. The frame's area is $(s + 4)^2 - s^2$.

What do $(s + 4)^2$ and s^2 represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q4/5

True or false?

For every real number x, $5 + 3(x - 2)^2 \geq 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 1

Apply the distributive property to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

Expand.

$3(2x + 5)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q1/5

Factor using the greatest common factor.

12y + 18

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q2/5

Which expression is equal to $4(a - 3) + 2a$?

A. $6a + 12$

B. $6a - 12$

C. $6a - 3$

D. $4a - 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q3/5

True or false?

$$2(x + y) = 2x + 2y$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q4/5

True or false?

$$3(x + 4) = 3x + 4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 2

Evaluate expressions for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $(x - 3)^2$ when $x = 7$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q1/5

Evaluate $100(1.05)^2$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q2/5

Evaluate $10 - (x - 3)^2$ when $x = 1$.

A. 8

B. -6

C. 14

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q3/5

True or false?

When $x = 3$, $(x - 3)^2 = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q4/5

True or false?

When $x = -2$, $x^2 = -4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 1

Use the formula for the sum of a finite geometric series • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Find the sum of the first 6 terms of 3 + 6 + 12 + ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q1/5

Use $S = a(1 - r^n) \div (1 - r)$ to find $2 + 1 + 0.5 + 0.25 + 0.125$.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q2/5

A geometric series has first term 5, ratio 3, and 4 terms.

What is its sum?

A. 405

B. 80

C. 200

D. 121

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q3/5

True or false?

$$1 + 2 + 4 + \ldots + 2^{9} = 2^{10} - 1$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q4**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q4/5

True or false?

The formula $a(1 - r^n) \div (1 - r)$ gives the sum of a geometric series with $r = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q5**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 2

Solve exponential growth models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

An investment is worth $1{,}000(1.05)^n$ dollars after n years.

Write the number of years to reach $2,000 as an expression with logarithms, then round it to the nearest tenth.

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q1/5

A town of 5,000 people grows 3% per year: $P = 5{,}000(1.03)^t$. When will it reach 8,000? Round to the nearest tenth of a year. (log means base 10.)

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q2/5

Solve $5 \cdot 2^t = 60$. (log means base 10.)

A. $t = \log 60 \div \log 10 \approx 1.78$

B. $t = 3$

C. $t = \log 12 \div \log 2 \approx 3.58$

D. $t = 12 \div 2 = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q3/5

True or false?

$\log(1.05^n) = n \cdot \log 1.05$ (log means base 10.)

A. True

B. False

True or false?

The solution of $2^x = 10$ is $x = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q5/5

Add and subtract polynomials and explain why the result is a polynomial

MAIN	Algebra I main question	**HSA-APR.A.1**
BACKWARD 1	Add and subtract linear expressions	**7.EE.A.1**
BACKWARD 2	Combine like terms to write equivalent expressions	**6.EE.A.4**
FORWARD 1	Use the Remainder Theorem	**HSA-APR.B.2**
FORWARD 2	Rewrite a quotient of polynomials as q(x) + r(x)/b(x)	**HSA-APR.D.6**

SET 9 • HSA-APR.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Add. Write the answer in standard form.

$(4x^2 - 3x + 5) + (2x^2 + 7x - 9)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q1/5

Subtract. Write the answer in standard form.

$(5y^3 + 2y - 1) - (y^3 - 4y^2 + 6y - 3)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q2/5

Which expression is equal to $(3a^2 - 2ab) - (a^2 + 5ab)$?

A. $2a^2 - 7ab$

B. $2a^2 - 3ab$

C. $2a^2 + 3ab$

D. $4a^2 + 3ab$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-M-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q3/5

Explain why the sum of two polynomials is always a polynomial.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q4/5

True or false?

The difference of two polynomials is always a polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q5/5

SET 9 • HSA-APR.A.1

BACKWARD 1

Add and subtract linear expressions • Grade 7 • Expressions & Equations • 7.EE.A.1

Add.

$(3x + 4) + (5x - 9)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q1/5

Subtract.

$(7a - 2) - (3a - 6)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q2/5

Subtract.

$(2y + 1) - (y + 5)$

A. $y - 6$

B. $3y - 4$

C. $y + 6$

D. $y - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$$-(x - 3) = -x + 3$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q4/5

True or false?

$$(4n + 2) - (n + 2) = 3n + 4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSA-APR.A.1

BACKWARD 2

Combine like terms to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.4

Simplify.

$4m + 2 + m$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q1/5

Are 2(3 + x) and 6 + 2x equivalent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q2/5

Which expression is equal to b + b + b + 4?

A. 7b

B. 3b + 4

C. 4b + 3

D. $b^3 + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q3/5

True or false?

$x + x = 2x$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q4/5

True or false?

$5y - y = 4y$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSA-APR.A.1

FORWARD 1

Use the Remainder Theorem • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find the remainder when $x^2 + 3x - 4$ is divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q1/5

Is $x + 1$ a factor of $x^3 + 1$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q2/5

What is the remainder when $p(x) = x^3 - 2x + 5$ is divided by $x - 1$?

A. 8

B. 4

C. -4

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q3/5

True or false?

If $p(3) = 0$, then $x - 3$ is a factor of $p(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q4/5

True or false?

The remainder when p(x) is divided by x + 2 is p(2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSA-APR.A.1

FORWARD 2

Rewrite a quotient of polynomials as q(x) + r(x)/b(x) • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Assume $x \neq -2$. Write $(x^2 + 5x + 7) \div (x + 2)$ in the form $q(x) + r/(x + 2)$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q1/5

Assume $x \neq 0$. Write $(3x + 7) \div x$ in the form $a + b/x$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q2/5

Assume $x \neq 1$. Which expression is equal to $(x^2 - 1) \div (x - 1)$?

A. $x + 1$

B. x

C. $x^2 + 1$

D. $x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q3/5

True or false?

Assume $x \neq 0$. $(2x + 1) \div x = 2 + 1/x$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q4/5

True or false?

Assume $x \neq -2$. $(x^2 + 4) \div (x + 2) = x + 2$

A. True

B. False

Multiply polynomials

MAIN	Algebra I main question	**HSA-APR.A.1**
BACKWARD 1	Apply the distributive property to linear expressions	**6.EE.A.3**
BACKWARD 2	Multiply powers with the same base	**8.EE.A.1**
FORWARD 1	Prove polynomial identities	**HSA-APR.C.4**
FORWARD 2	Expand powers of binomials with the binomial theorem	**HSA-APR.C.5**

SET 10 • HSA-APR.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply. Write the answer in standard form.

$(2x + 3)(x - 5)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q1/5

Multiply. Write the answer in standard form.

$(x - 4)^2$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q2/5

Multiply. Write the answer in standard form.

$(y + 2)(y^2 - 3y + 1)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q3/5

Multiply.

$(3a - 1)(3a + 1)$

A. $9a^2 - 1$

B. $9a^2 - 6a - 1$

C. $6a - 1$

D. $9a^2 + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q4/5

True or false?

$(x + 3)^2 = x^2 + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q5/5

SET 10 • HSA-APR.A.1

BACKWARD 1

Apply the distributive property to linear expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

Expand.

$4(3x - 2)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q1/5

Expand.

$-2(5 - 3a)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q2/5

Expand.

$7(2b + 1)$

A. $14b + 8$

B. $14b + 7$

C. $9b + 8$

D. $14b + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q3/5

True or false?

$$5(m - 2) = 5m - 10$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q4/5

True or false?

$$3(2x + 4) = 6x + 4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSA-APR.A.1

BACKWARD 2

Multiply powers with the same base • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$7^2 \cdot 7^3$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q1/5

Write as a single power.

$10^4 \cdot 10$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q2/5

Write as a single power.

$6^3 \cdot 6^3$

A. 6^6

B. 6^9

C. 12^6

D. 36^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q3/5

True or false?

$9 \cdot 9^4 = 9^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q4/5

True or false?

$2^3 \cdot 2^3 = 2^9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSA-APR.A.1

FORWARD 1

Prove polynomial identities • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Show that $(a + b)^2 - (a - b)^2 = 4ab$ for all numbers a and b.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q1/5

Use the identity $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$ to factor $x^3 - 8$.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q2/5

Which identity is true for all numbers a and b?

A. $(a - b)^2 = a^2 - b^2$

B. $(a + b)(a - b) = a^2 - b^2$

C. $(a + b)^2 = a^2 + b^2$

D. $a^2 + b^2 = (a + b)(a + b)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q3/5

True or false?

$(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ for all x and y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q4/5

True or false?

$(x + 1)^3 = x^3 + 1$ for all x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSA-APR.A.1

FORWARD 2

Expand powers of binomials with the binomial theorem • High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Expand. Write the answer in standard form.

$(2x - 1)^3$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q1/5

What is the coefficient of the x^3y^2 term in the expansion of $(x + y)^5$?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q2/5

How many terms does the expansion of $(a + b)^7$ have?

A. 128

B. 8

C. 14

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q3/5

True or false?

The number 5 choose 2 equals 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q4/5

True or false?

Every coefficient in the expansion of $(a - b)^4$ is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q5/5

Use structure to factor with a greatest common factor and the difference of squares

MAIN	Algebra I main question	**HSA-SSE.A.2**
BACKWARD 1	Use the greatest common factor and the distributive property	**6.NS.B.4**
BACKWARD 2	Multiply a sum and a difference	**HSA-APR.A.1**
FORWARD 1	Find the zeros of a polynomial from its factors	**HSA-APR.B.3**
FORWARD 2	Factor sums of squares with complex numbers	**HSN-CN.C.8**

SET 11 • HSA-SSE.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor completely.

$6x^3 - 15x^2$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q1/5

Factor completely.

$49y^2 - 25$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q2/5

Factor completely.

$x^4 - 16$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q3/5

Which is the complete factorization of $2x^2 - 18$?

A. $(2x - 6)(x + 3)$

B. $2(x - 3)(x + 3)$

C. $(x - 3)(x + 3)$

D. $2(x^2 - 9)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q4/5

True or false?

$x^2 + 9$ can be factored as $(x + 3)(x - 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q5/5

SET 11 • HSA-SSE.A.2

BACKWARD 1

Use the greatest common factor and the distributive property • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 36.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q1/5

Use the greatest common factor to write 42 + 28 as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q2/5

What is the greatest common factor of 15 and 40?

A. 5

B. 8

C. 120

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q3/5

True or false?

$$18 + 12 = 6(3 + 2)$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q4/5

True or false?

The greatest common factor of 9 and 16 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSA-SSE.A.2

BACKWARD 2

Multiply a sum and a difference • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply.

$(x - 7)(x + 7)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q1/5

Multiply.

$(2y + 3)(2y - 3)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q2/5

Multiply.

$(x + 4)(x - 4)$

A. $x^2 - 8$

B. $x^2 + 16$

C. $x^2 - 8x - 16$

D. $x^2 - 16$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-B2-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q3/5

True or false?

$$(a - 5)(a + 5) = a^2 - 25$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q4/5

True or false?

$$(x + 3)(x - 3) = x^2 - 6x - 9$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSA-SSE.A.2

FORWARD 1

Find the zeros of a polynomial from its factors • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

What are the zeros of $p(x) = x(x - 2)(x + 5)$?

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions HSA-APR.B.3 • Set 11 • FORWARD 1 • Q1/5

Factor $x^3 - 9x$ completely, then give all of its zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q2/5

Which polynomial has the zeros -1, 0, and 4?

A. $(x + 1)(x - 4)$

B. $x(x + 1)(x + 4)$

C. $x(x + 1)(x - 4)$

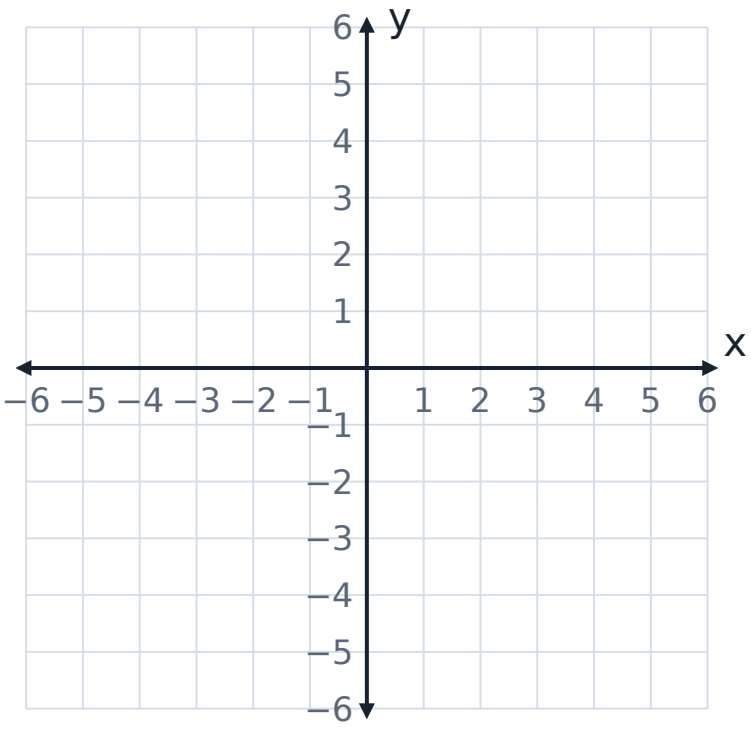
D. $x(x - 1)(x + 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q3/5

Sketch a rough graph of $p(x) = x(x - 2)(x + 5)$. Mark the zeros and show the end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions HSA-APR.B.3 • Set 11 • FORWARD 1 • Q4/5

True or false?

$p(x) = (x - 3)^2(x + 1)$ has three different zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSA-SSE.A.2

FORWARD 2

Factor sums of squares with complex numbers • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^2 + 25$ using complex numbers.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q1/5

Solve $x^2 + 16 = 0$. Give all solutions.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q2/5

Which is the complete factorization of $x^4 - 16$ into linear factors with complex numbers?

A. $(x - 4)(x + 4)$

B. $(x^2 - 4)(x^2 + 4)$

C. $(x - 2)(x + 2)(x - 2i)(x + 2i)$

D. $(x - 2)^2(x + 2)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q3/5

True or false?

$$(x + i)(x - i) = x^2 + 1$$

A. True

B. False

True or false?

$x^2 + 4$ has no linear factors, even with complex numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q5/5

Factor quadratic trinomials

MAIN	Algebra I main question	**HSA-SSE.A.2**
BACKWARD 1	Find factor pairs of a whole number	**4.OA.B.4**
BACKWARD 2	Multiply two binomials	**HSA-APR.A.1**
FORWARD 1	Use factored form to describe the graph of a polynomial function	**HSF-IF.C.7.c**
FORWARD 2	Write a polynomial from its zeros and find zeros by factoring	**HSA-APR.B.3**

SET 12 • HSA-SSE.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor.

$x^2 + 7x + 12$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q1/5

Factor.

$x^2 - 2x - 15$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q2/5

Factor completely.

$2x^2 + 7x + 3$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q3/5

Which is a factor of $3x^2 - 10x - 8$?

A. $3x - 2$

B. $x + 4$

C. $x - 4$

D. $x - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q4/5

True or false?

$x^2 + 5x + 6 = (x + 1)(x + 6)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q5/5

SET 12 • HSA-SSE.A.2

BACKWARD 1

Find factor pairs of a whole number • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factor pairs of 12.

Factor pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q1/5

Two whole numbers multiply to 15 and add to 8.

What are they?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q2/5

Which factor pair of 24 has a sum of 10?

A. 4 and 6

B. 3 and 8

C. 1 and 24

D. 2 and 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q3/5

True or false?

3 and 7 are a factor pair of 21.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q4/5

True or false?

4 and 5 are a factor pair of 25.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSA-SSE.A.2

BACKWARD 2

Multiply two binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply.

$(x + 2)(x + 9)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q1/5

Multiply.

$(x - 6)(x + 1)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q2/5

Multiply.

$(3x + 1)(x + 2)$

A. $3x^2 + 6x + 2$

B. $3x^2 + 7x + 2$

C. $4x^2 + 7x + 2$

D. $3x^2 + 3x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-B2-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q3/5

True or false?

$$(x - 2)(x - 6) = x^2 - 8x + 12$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q4/5

True or false?

$$(x + 1)(x - 4) = x^2 - 3x + 4$$

A. True

B. False

SET 12 • HSA-SSE.A.2

FORWARD 1

Use factored form to describe the graph of a polynomial function • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

$p(x) = (x + 2)(x - 1)(x - 3)$.

What are its zeros and its y-intercept?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q1/5

Describe the end behavior of $y = x^3 - 4x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q2/5

Which describes the end behavior of $y = -x^3$?

A. It rises to the left and falls to the right.

B. It rises on both ends.

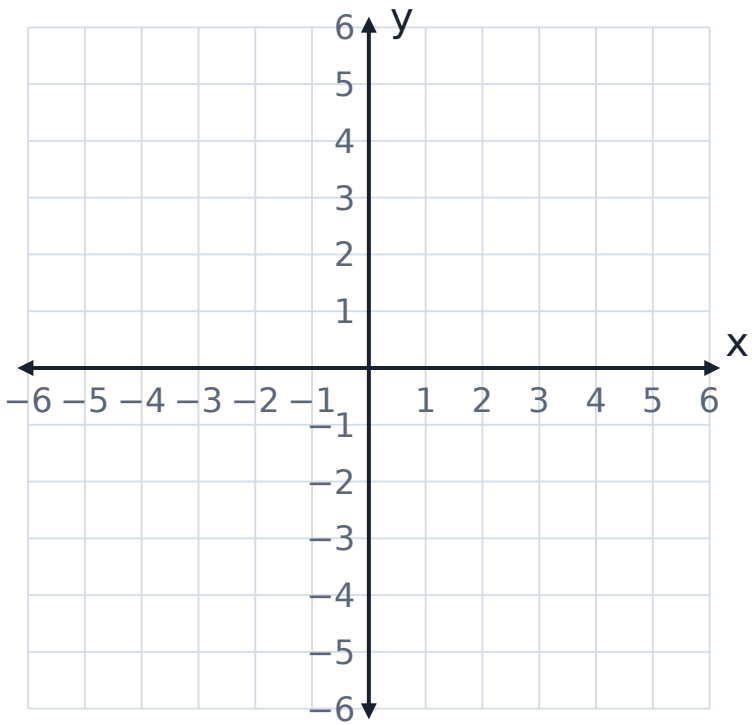
C. It falls to the left and rises to the right.

D. It falls on both ends.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q3/5

Graph
$p(x) = (x + 2)(x - 1)(x - 3)$.
Mark the zeros and the y-intercept, and show the end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q4/5

True or false?

The graph of $y = (x - 1)^2$ crosses the x-axis at $x = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSA-SSE.A.2

FORWARD 2

Write a polynomial from its zeros and find zeros by factoring • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Write a polynomial function of degree 3 whose only zeros are 2, −3, and 5.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q1/5

Factor $x^3 - x^2 - 6x$ completely, then give its zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q2/5

What are the zeros of $p(x) = (2x - 1)(x + 4)$?

A. 2 and -4

B. 1 and 4

C. $\frac{1}{2}$ and -4

D. $-\frac{1}{2}$ and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q3/5

True or false?

If $(x - 7)$ is a factor of $p(x)$, then 7 is a zero of p.

A. True

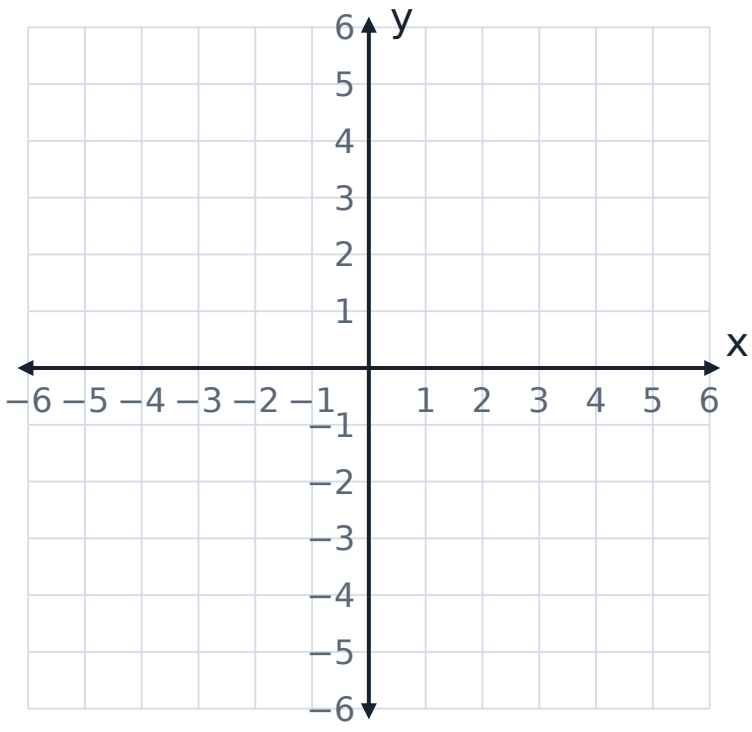
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q4/5

Sketch a rough graph of $p(x) = (x - 1)^2(x + 2)(x - 3)$. Show what the graph does at each zero and its end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q5/5

Use structure to factor perfect squares and expressions in quadratic form

MAIN	Algebra I main question	**HSA-SSE.A.2**
BACKWARD 1	Find squares of whole numbers and decimals	**6.EE.A.1**
BACKWARD 2	Square a binomial	**HSA-APR.A.1**
FORWARD 1	Find the zeros of polynomial functions in quadratic form	**HSF-IF.C.7.c**
FORWARD 2	Factor quadratic expressions with complex numbers	**HSN-CN.C.8**

SET 13 • HSA-SSE.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor.

$x^2 + 10x + 25$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q1/5

Factor completely. (Think of x^2 as one quantity.)

$x^4 - 5x^2 + 4$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q2/5

Factor, and simplify each factor. (Think of x + 1 as one quantity.)

$(x + 1)^2 - 9$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q3/5

Which expression is a perfect square trinomial?

A. $x^2 + 9$

B. $4x^2 - 12x + 9$

C. $4x^2 + 6x + 9$

D. $x^2 - 6x - 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q4/5

True or false?

$9x^2 + 30x + 25 = (3x + 5)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q5/5

SET 13 • HSA-SSE.A.2

BACKWARD 1

Find squares of whole numbers and decimals • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

15^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q1/5

What whole number squared is 169?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q2/5

Find the value.

2.5^2

A. 5

B. 4.25

C. 62.5

D. 6.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q3/5

True or false?

$11^2 = 121$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q4/5

True or false?

$6^2 = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSA-SSE.A.2

BACKWARD 2

Square a binomial • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply.

$(x + 5)^2$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q1/5

Multiply.

$(3y - 2)^2$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q2/5

Multiply.

$(x - 6)^2$

A. $x^2 - 6x + 36$

B. $x^2 - 36$

C. $x^2 + 36$

D. $x^2 - 12x + 36$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-B2-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q3/5

True or false?

$(a + 1)^2 = a^2 + 2a + 1$

A. True

B. False

True or false?

$(x - 3)^2 = x^2 - 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSA-SSE.A.2

FORWARD 1

Find the zeros of polynomial functions in quadratic form • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

Find all real zeros of $f(x) = x^4 - 5x^2 + 4$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q1/5

Find all real zeros of $y = x^4 - 9x^2$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q2/5

How many real zeros does $y = x^4 - 1$ have?

A. 2

B. 1

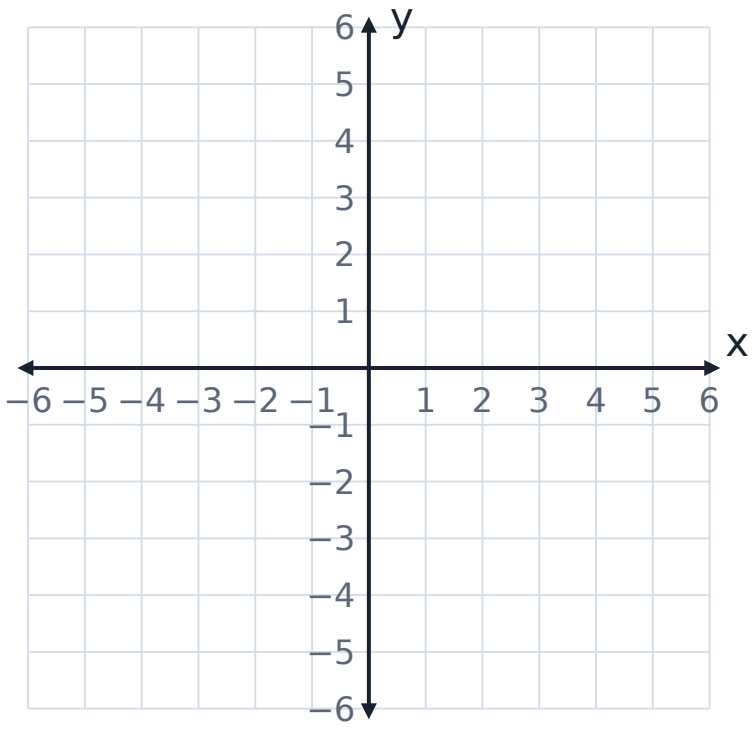
C. 0

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q3/5

Graph $f(x) = x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4)$. Mark the zeros and the y-intercept, and show the end behavior.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q4/5

True or false?

$y = x^4 + 1$ has a real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSA-SSE.A.2

FORWARD 2

Factor quadratic expressions with complex numbers • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^2 + 49$ using complex numbers.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q1/5

The solutions of $x^2 + 2x + 5 = 0$ are $-1 + 2i$ and $-1 - 2i$.

Write $x^2 + 2x + 5$ as a product of two linear factors.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q2/5

Multiply.

$(x - 3i)(x + 3i)$

A. $x^2 - 9$

B. $x^2 + 6ix - 9$

C. $x^2 + 9$

D. $x^2 - 9i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q3/5

True or false?

$$x^4 - 1 = (x - 1)(x + 1)(x - i)(x + i)$$

A. True

B. False

True or false?

$(x + 2i)^2 = x^2 - 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q5/5

Factor a quadratic expression to reveal the zeros of the function it defines

MAIN	Algebra I main question	**HSA-SSE.B.3.a**
BACKWARD 1	Factor trinomials with a leading coefficient of 1	**HSA-SSE.A.2**
BACKWARD 2	Use substitution to decide which values make an equation true	**6.EE.B.5**
FORWARD 1	Find zeros of cubic functions by factoring	**HSA-APR.B.3**
FORWARD 2	Sketch polynomial graphs from their zeros	**HSF-IF.C.7.c**

SET 14 • HSA-SSE.B.3.a

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

$f(x) = x^2 - x - 12$.

Factor f(x), then give the zeros of f.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q1/5

A ball's height is $h(t) = -16t^2 + 48t$ feet after t seconds.

Factor h(t), and use the factors to say when the ball is at ground level.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q2/5

What are the zeros of $g(x) = 2x^2 + 5x - 3$?

A. $\frac{1}{2}$ and -3

B. $-\frac{1}{2}$ and 3

C. 2 and -3

D. 1 and -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q3/5

Write a quadratic function, in factored form with a leading coefficient of 1, whose zeros are 2 and −5.

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q4/5

True or false?

The zeros of $f(x) = (x - 6)(x + 1)$ are 6 and -1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q5/5

SET 14 • HSA-SSE.B.3.a

BACKWARD 1

Factor trinomials with a leading coefficient of 1 • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor.

$x^2 + 9x + 20$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q1/5

Factor.

$x^2 - 9x + 14$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q2/5

Factor.

$x^2 + x - 6$

A. $(x - 6)(x + 1)$

B. $(x - 3)(x + 2)$

C. $(x + 3)(x - 2)$

D. $(x + 6)(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q3/5

True or false?

$$x^2 - 4x - 21 = (x - 7)(x + 3)$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q4/5

True or false?

$x^2 + 8x + 15 = (x + 3)(x + 6)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSA-SSE.B.3.a

BACKWARD 2

Use substitution to decide which values make an equation true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is $x = 4$ a solution of $x^2 = 16$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q1/5

Which of the values 1, 2, and 3 make $x^2 - 5x + 6 = 0$ true?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q2/5

Which value makes $3x - 6 = 0$ true?

A. 3

B. -2

C. 2

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q3/5

True or false?

$x = 5$ makes $x(x - 5) = 0$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q4/5

True or false?

$x = -2$ makes $x + 2 = 4$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSA-SSE.B.3.a

FORWARD 1

Find zeros of cubic functions by factoring • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Find the zeros of $f(x) = x^3 + 2x^2 - 15x$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q1/5

$g(x) = (x - 1)(x^2 - 4)$.

List all the zeros of g.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q2/5

Which function has a double zero at $x = 2$?

A. $f(x) = (x - 2)^2(x + 1)$

B. $f(x) = 2x(x - 1)$

C. $f(x) = (x + 2)^2(x - 1)$

D. $f(x) = (x - 2)(x + 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q3/5

True or false?

$f(x) = x^3 - x$ has three real zeros.

A. True

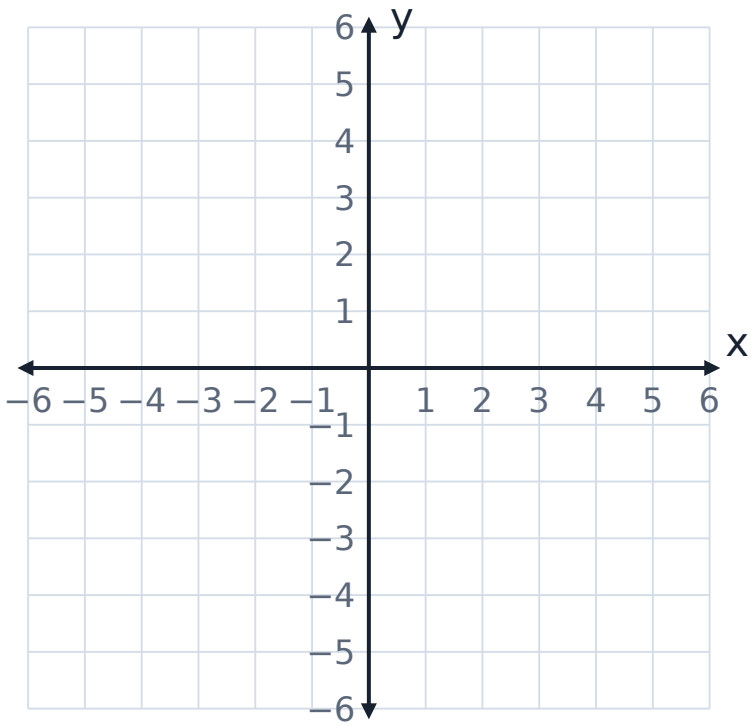
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q4/5

Sketch a rough graph of $f(x) = (x - 2)^2(x + 1)$. Mark the zeros and the y-intercept, and show what the graph does at each zero. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSA-SSE.B.3.a

FORWARD 2

Sketch polynomial graphs from their zeros • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

Describe where $f(x) = (x + 1)(x - 2)(x - 4)$ is above the x-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q1/5

A cubic function has zeros -3, 0, and 3 and a positive leading coefficient.

Describe its end behavior.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q2/5

How many turning points can the graph of a cubic function have, at most?

A. 2

B. 1

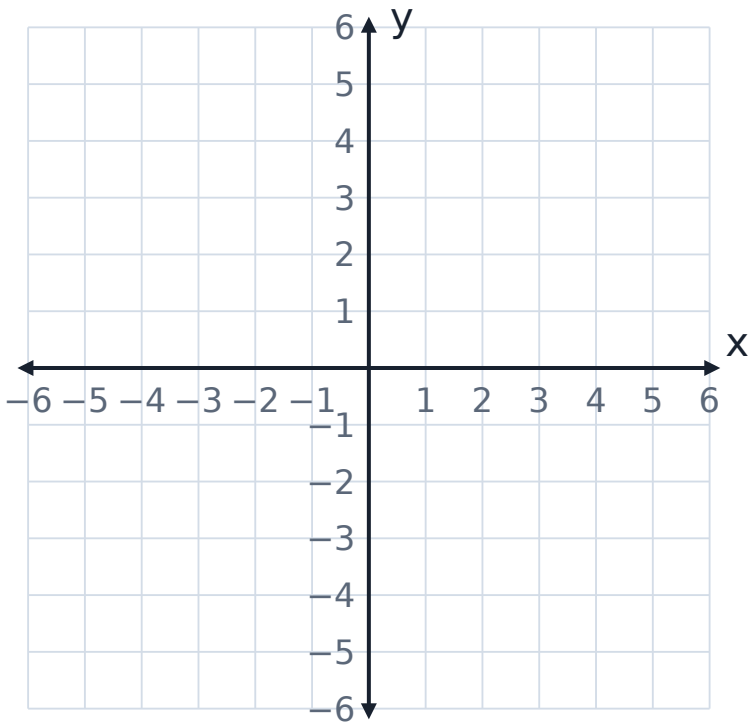
C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q3/5

Graph
$f(x) = (x + 1)(x - 2)(x - 4)$.
Mark the zeros and the y-intercept, and show the end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q4/5

True or false?

A polynomial function with a negative leading coefficient always falls on both ends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q5/5

Complete the square to reveal the maximum or minimum value of a quadratic function

MAIN	Algebra I main question	**HSA-SSE.B.3.b**
BACKWARD 1	Recognize and complete perfect square trinomials	**HSA-SSE.A.2**
BACKWARD 2	Describe where a graph rises, falls, and reaches its highest point	**8.F.B.5**
FORWARD 1	Complete the square to find the center and radius of a circle	**HSG-GPE.A.1**
FORWARD 2	Derive the equations of ellipses and hyperbolas from their foci	**HSG-GPE.A.3**

SET 15 • HSA-SSE.B.3.b

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

Write $f(x) = x^2 + 6x + 5$ in vertex form by completing the square. Show the steps.

$f(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q1/5

$g(x) = x^2 - 10x + 30$.

Complete the square to write g in vertex form, then give the minimum value of g and where it happens. Show the steps.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q2/5

A ball's height is $h(t) = -t^2 + 8t + 2$ meters after t seconds.

Complete the square (write h in vertex form) to find the greatest height and when it happens. Show the steps.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q3/5

Which is $x^2 - 4x + 7$ in vertex form?

A. $(x - 4)^2 + 7$

B. $(x - 2)^2 + 7$

C. $(x - 2)^2 + 3$

D. $(x + 2)^2 + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q4/5

True or false?

The minimum value of $(x + 1)^2 - 6$ is -6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q5/5

SET 15 • HSA-SSE.B.3.b

BACKWARD 1

Recognize and complete perfect square trinomials • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor.

$x^2 + 14x + 49$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q1/5

What number c makes $x^2 - 8x + c$ a perfect square trinomial?

$c =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q2/5

Which number completes $x^2 + 12x + ___$ to a perfect square trinomial?

A. 36

B. 6

C. 24

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q3/5

True or false?

$$x^2 - 2x + 1 = (x - 1)^2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q4/5

True or false?

$x^2 + 6x + 6$ is a perfect square trinomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

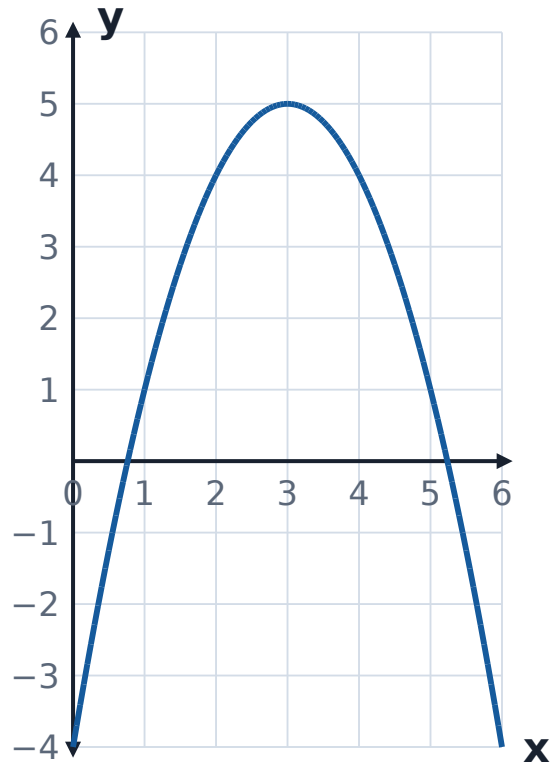
High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSA-SSE.B.3.b

BACKWARD 2

Describe where a graph rises, falls, and reaches its highest point • Grade 8 • Functions • 8.F.B.5

Where is the highest point of the graph?



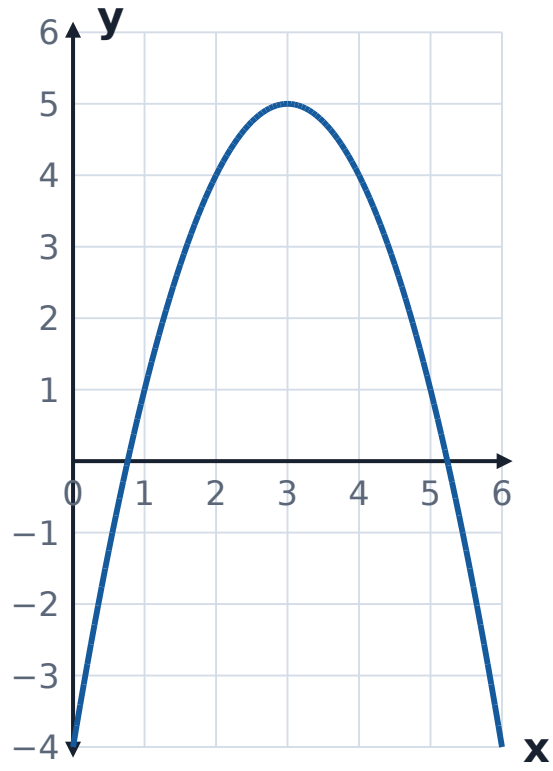
Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q1/5

For which x-values shown is the graph rising?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q2/5

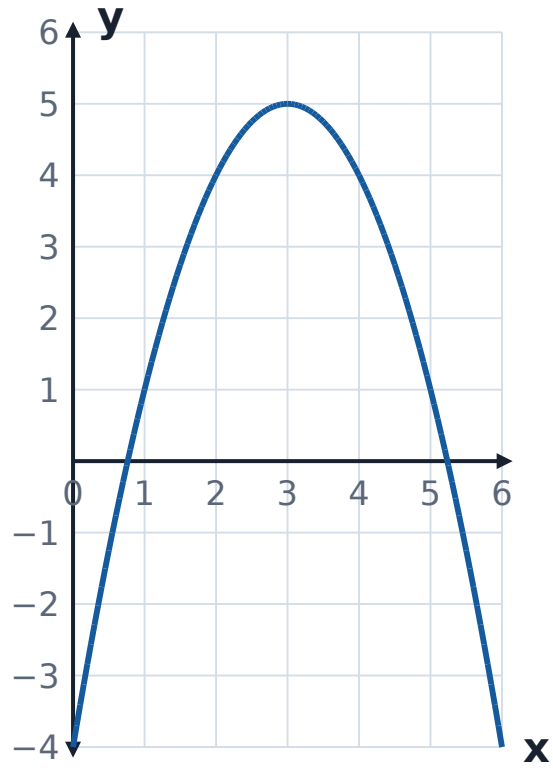
For which x-values shown is the graph falling?

A. From x = 3 to x = 6

B. It never falls.

C. From x = 0 to x = 3

D. From x = 0 to x = 6



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

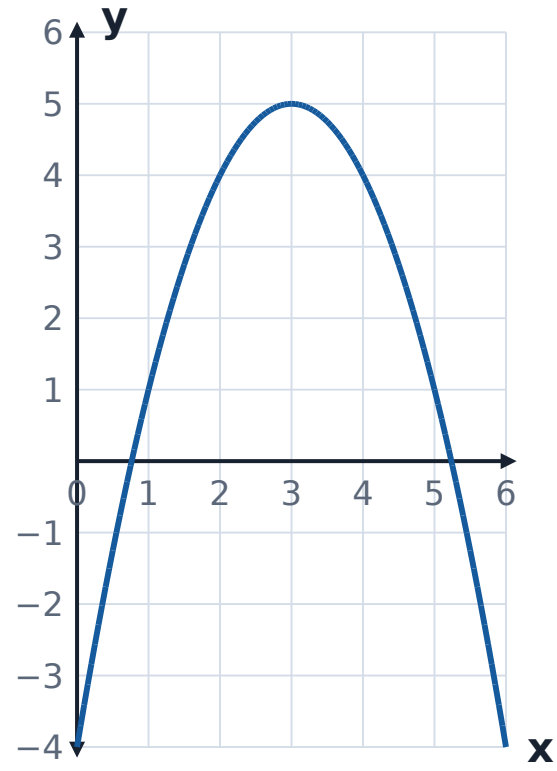
Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q3/5

True or false?

The graph has the same height at x = 2 and at x = 4.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

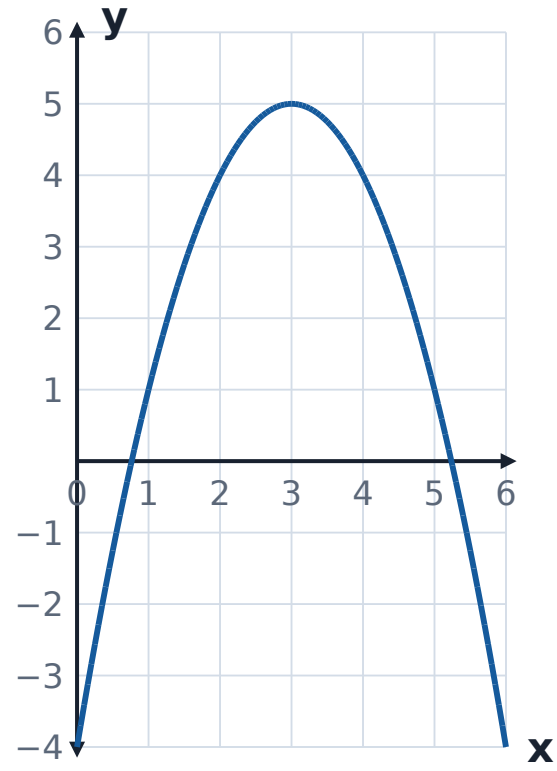
Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q4/5

True or false?

The graph is a straight line.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSA-SSE.B.3.b

FORWARD 1

Complete the square to find the center and radius of a circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Find the center and radius of the circle $x^2 + y^2 - 6x + 4y - 3 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q1/5

Write an equation of the circle with center $(-1, 4)$ and radius 3 in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q2/5

Find the center and radius of $x^2 + y^2 + 8x = 9$.

A. Center (4, 0), radius 5

B. Center (0, 0), radius 9

C. Center (−4, 0), radius 3

D. Center (−4, 0), radius 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q3/5

True or false?

The circle $(x - 2)^2 + (y + 1)^2 = 4$ has a radius of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q4/5

True or false?

The circle $x^2 + y^2 = 25$ has its center at (5, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSA-SSE.B.3.b

FORWARD 2

Derive the equations of ellipses and hyperbolas from their foci • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

An ellipse has foci $(4, 0)$ and $(-4, 0)$, and the sum of the distances from any point on it to the foci is 10.

Write its equation in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q1/5

A hyperbola has foci (5, 0) and (−5, 0), and the absolute value of the difference of the distances from any point on it to the foci is 8.

Write its equation in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q2/5

The ellipse $x^2 \div 25 + y^2 \div 9 = 1$ has foci $(4, 0)$ and $(-4, 0)$.

Show that $(3, 2.4)$ is on the ellipse and that its distances to the two foci add to 10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q3/5

True or false?

$x^2/25 + y^2/9 = 1$ is the equation of an ellipse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q4/5

True or false?

For every point on an ellipse, the difference of the distances to the two foci is constant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q5/5

Use the properties of exponents to transform exponential expressions

MAIN	Algebra I main question	**HSA-SSE.B.3.c**
BACKWARD 1	Use the meaning of rational exponents and the power-of-a-power…	**HSN-RN.A.1**
BACKWARD 2	Solve percent increase and decrease problems	**7.RP.A.3**
FORWARD 1	Solve exponential equations with logarithms	**HSF-LE.A.4**
FORWARD 2	Use the sum of a geometric series in a growth model	**HSA-SSE.B.4**

SET 16 • HSA-SSE.B.3.c

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c

An account grows by 6% each year: $A = 500(1.06)^t$ after t years.

Rewrite the expression to show the monthly growth factor, rounded to four decimal places.

$A =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q1/5

Use the properties of exponents to explain why $1.21^{t} = 1.1^{2t}$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q2/5

A population doubles every 5 years: $P = 300 \cdot 2^{t/5}$ after t years.

Find the yearly growth factor to three decimal places, and the yearly percent growth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q3/5

Which expression is equal to 8^t?

A. 2^{t+3}

B. 3^{2t}

C. 4^{2t}

D. 2^{3t}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q4/5

True or false?

In $y = 100(0.5)^{t/2}$, the amount is cut in half every 2 time units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q5/5

SET 16 • HSA-SSE.B.3.c

BACKWARD 1

Use the meaning of rational exponents and the power-of-a-power property • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

Find the value.

$4^{1/2}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q1/5

Write $(1.1^{2})^{t}$ as a single power of 1.1.

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q2/5

Find the value.

$9^{1/2}$

A. $\frac{1}{9}$

B. 4.5

C. 3

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q3/5

True or false?

$(2^3)^t = 2^{3t}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q4/5

True or false?

$5^{1/2} = 2.5$

A. True

B. False

SET 16 • HSA-SSE.B.3.c

BACKWARD 2

Solve percent increase and decrease problems • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Increase $500 by 6%.

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q1/5

Decrease 80 by 25%.

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q2/5

What do you multiply by to increase an amount by 4%?

A. 0.04

B. 1.4

C. 1.04

D. 0.96

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q3/5

True or false?

Decreasing an amount by 10% is the same as multiplying it by 0.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q4/5

True or false?

An increase of 5% followed by a decrease of 5% gives back the original amount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSA-SSE.B.3.c

FORWARD 1

Solve exponential equations with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $500(1.06)^t = 800$. Round t to the nearest tenth.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q1/5

A population is $P = 300 \cdot 2^{t/5}$ after t years. When is $P = 1{,}000$? Round to the nearest tenth. (log means base 10.)

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q2/5

Solve $10^{2x} = 50$. (log means base 10.)

A. $x = 2.5$

B. $x = (\log 50) \div 2 \approx 0.85$

C. $x = 25$

D. $x = \log 25 \approx 1.40$

True or false?

If $3^x = 3^5$, then x = 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q4/5

True or false?

The solution of $1.06^t = 2$ is $t = 2 \div 1.06$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSA-SSE.B.3.c

FORWARD 2

Use the sum of a geometric series in a growth model • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Deposits of $100, $100(1.005), $100(1.005)^2, ..., $100(1.005)^{11} are added. Use $S = a(1 - r^n) \div (1 - r)$ to find the total, to the nearest dollar.

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q1/5

Find the sum 5 + 10 + 20 + ... + 640.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q2/5

What is $1 + 3 + 9 + 27 + 81$?

A. 81

B. 121

C. 243

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q3/5

True or false?

The series 2 + 6 + 18 + 54 has a common ratio of 3.

A. True

B. False

True or false?

The sum of the first n terms of a geometric series with first term a and ratio $r \neq 1$ is $a \cdot r^n$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q5/5

Explain each step in solving an equation and justify the solution method

MAIN	Algebra I main question	**HSA-REI.A.1**
BACKWARD 1	Use substitution to decide whether a number makes an equation…	**6.EE.B.5**
BACKWARD 2	Solve linear equations by expanding and collecting like terms	**8.EE.C.7.b**
FORWARD 1	Justify the steps of a proof about lines and angles	**HSG-CO.C.9**
FORWARD 2	Solve radical and rational equations and identify extraneous solut..	**HSA-REI.A.2**

SET 17 • HSA-REI.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

Solve $3(x - 4) + 5 = 2x + 7$. Give a reason for each step.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q1/5

A student solves $5x - 8 = 12$.

Step 1: $5x = 20$

Step 2: $x = 4$

Which property justifies Step 1?

A. Distributive property

B. Addition property of equality: add 8 to both sides

C. Subtraction property of equality: subtract 8 from both sides

D. Multiplication property of equality: multiply both sides by 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q2/5

A student wrote:

$4(x + 2) = 20$

$4x + 2 = 20$

$4x = 18$

$x = 4.5$

Which step has the error? What is the correct solution?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q3/5

True or false?

If $a = b$, then $a \div c = b \div c$ for every number c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q4/5

True or false?

Subtracting 2x from both sides of $2(x + 1) = 2x + 2$ gives $2 = 2$, so every real number is a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q5/5

SET 17 • HSA-REI.A.1

BACKWARD 1

Use substitution to decide whether a number makes an equation true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is x = 3 a solution of 4x − 5 = 7? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q1/5

Which number from 1, 2, 3, 4 makes 2x + 1 = 9 true?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q2/5

Which value of y makes $15 - y = 9$ true?

A. 6

B. -6

C. 9

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q3/5

True or false?

$x = 5$ is a solution of $3x = 15$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q4/5

True or false?

$n = 2$ is a solution of $n + 7 = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSA-REI.A.1

BACKWARD 2

Solve linear equations by expanding and collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve $2(x + 3) = 14$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q1/5

Solve $5y - 2y + 4 = 19$.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q2/5

Solve $3(n - 2) = n + 8$.

A. $n = 7$

B. $n = 5$

C. $n = 3$

D. $n = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q3/5

True or false?

The solution of $0.5x + 1 = 4$ is $x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q4/5

True or false?

The solution of $4(x - 1) = 12$ is $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSA-REI.A.1

FORWARD 1

Justify the steps of a proof about lines and angles • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9

Angles 1 and 2 are vertical angles. Angle 1 and angle 3 form a linear pair, and so do angle 2 and angle 3.

Explain why $m\angle 1 = m\angle 2$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q1/5

In a proof, the line $m\angle A + 40° = 90°$ is followed by $m\angle A = 50°$.

Which reason justifies that step?

A. Definition of a linear pair

B. Vertical angles are congruent

C. Addition property of equality

D. Subtraction property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q2/5

Two angles form a linear pair. One measures $(3x + 10)°$ and the other measures $(2x + 20)°$.

Find x and both angle measures.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q3/5

True or false?

Vertical angles have equal measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q4/5

True or false?

If two angles have equal measures, they must be vertical angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSA-REI.A.1

FORWARD 2

Solve radical and rational equations and identify extraneous solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $\sqrt{(x + 3)} = 5$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q1/5

Solve $\sqrt{(2x + 3)} = x$. Name any extraneous solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q2/5

Assume $x \neq 0$. Solve $6 \div x = 3$.

A. $x = \frac{1}{2}$

B. $x = 3$

C. $x = 18$

D. $x = 2$

True or false?

$x = 4$ is a solution of $\sqrt{x} = -2$.

A. True

B. False

True or false?

Squaring both sides of an equation can produce a value that does not satisfy the original equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q5/5

Solve linear equations in one variable, including equations with letter coefficients

MAIN	Algebra I main question	**HSA-REI.B.3**
BACKWARD 1	Name the property that justifies a step	**HSA-REI.A.1**
BACKWARD 2	Solve two-step equations from word problems	**7.EE.B.4.a**
FORWARD 1	Prove and apply theorems about triangles	**HSG-CO.C.10**
FORWARD 2	Solve rational equations that lead to linear equations	**HSA-REI.A.2**

SET 18 • HSA-REI.B.3

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $\frac{2}{3}x - 5 = \frac{1}{4}x + 5$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q1/5

Solve $0.4(3x - 5) = 2.2x + 1$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q2/5

How many solutions does $5x - 2(x + 4) = 3x + 1$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q3/5

Solve $ax + b = c$ for x, where $a \neq 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q4/5

True or false?

The equation $2(x - 3) = 2x - 6$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q5/5

SET 18 • HSA-REI.B.3

BACKWARD 1

Name the property that justifies a step • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

Name the property used to go from $3x + 4 = 19$ to $3x = 15$.

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q1/5

Name the property used to go from $\frac{x}{5} = 3$ to $x = 15$.

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q2/5

Which step keeps the same solutions as 2x = 10?

A. Divide both sides by 2

B. Square the right side only

C. Add 2 to the left side only

D. Multiply both sides by 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q3/5

True or false?

Adding the same number to both sides of an equation keeps the same solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q4/5

True or false?

Multiplying both sides of an equation by 0 keeps the same solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSA-REI.B.3

BACKWARD 2

Solve two-step equations from word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A taxi charges $3 plus $2 per mile. A ride cost $19.

How many miles was the ride?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q1/5

Solve $-4x + 9 = 1$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q2/5

Solve $\frac{1}{2}(x + 6) = 10$.

A. $x = 14$

B. $x = 17$

C. $x = 2$

D. $x = 26$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q3/5

True or false?

The solution of $6x - 4 = 20$ is $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q4/5

True or false?

The solution of $2x + 5 = -1$ is $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSA-REI.B.3

FORWARD 1

Prove and apply theorems about triangles • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10

Prove that the angles of any triangle ABC add to 180°. Start by drawing the line through B parallel to AC.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q1/5

Explain why an exterior angle of a triangle equals the sum of the two remote interior angles.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q2/5

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C. (Draw the bisector of ∠A, meeting BC at D.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q3/5

D and E are the midpoints of AB and AC in triangle ABC, and BC = 18.

What does the midsegment theorem let you conclude?

A. DE ⊥ BC and DE = 9

B. DE ∥ BC and DE = 18

C. DE = 9, but DE need not be parallel to BC

D. DE ∥ BC and DE = 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q4/5

True or false?

The three medians of a triangle meet at one point, and that point is $\frac{2}{3}$ of the way from each vertex to the midpoint of the opposite side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSA-REI.B.3

FORWARD 2

Solve rational equations that lead to linear equations • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Assume $x \neq 0$. Solve $3 \div x + 2 = 5 \div x$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q1/5

Assume $x \neq 2$. Solve $(x + 1) \div (x - 2) = 4$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q2/5

Assume $x \neq 3$. Solve $x \div (x - 3) = 3 \div (x - 3)$.

A. $x = 0$

B. No solution: $x = 3$ is extraneous

C. $x = 3$

D. Every real number except 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q3/5

True or false?

$x = 1$ is a solution of $2x \div (x + 1) = 1$.

A. True

B. False

True or false?

The solution of $6 \div (x + 1) = 2$ is $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q5/5

Solve linear inequalities and compound inequalities in one variable

MAIN	Algebra I main question	**HSA-REI.B.3**
BACKWARD 1	Solve two-step inequalities from word problems	**7.EE.B.4.b**
BACKWARD 2	Write and graph simple inequalities	**6.EE.B.8**
FORWARD 1	Use inequalities to meet constraints in a design problem	**HSG-MG.A.3**
FORWARD 2	Write a margin-of-error interval as a compound inequality — neare…	**HSS-IC.B.4**

SET 19 • HSA-REI.B.3

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $5 - 3x \leq 17$.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q1/5

Solve $-2 < 3x + 4 \leq 13$.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q2/5

The number line shows the solution set of which inequality?



A. $x + 5 > 7$

B. $4x + 3 < 2x + 7$

C. $3x \leq 6$

D. $-2x < -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q3/5

Solve $\frac{x}{4} - 1 > \frac{x}{2} + 2$.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q4/5

True or false?

The solution of $-4x > 8$ is $x > -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q5/5

SET 19 • HSA-REI.B.3

BACKWARD 1

Solve two-step inequalities from word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.b

Solve $3x + 2 > 14$.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q1/5

A phone plan costs $20 plus $5 per gigabyte g. Lena can spend at most $45.

Write and solve an inequality for g.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q2/5

Solve $-2x + 1 \geq 9$.

A. $x \leq -4$

B. $x \geq 5$

C. $x \leq 4$

D. $x \geq -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q3/5

True or false?

$x = 6$ is a solution of $2x - 3 > 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q4/5

True or false?

Dividing both sides of an inequality by a positive number reverses the inequality sign.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSA-REI.B.3

BACKWARD 2

Write and graph simple inequalities • Grade 6 • Expressions & Equations • 6.EE.B.8

A rider must be at least 48 inches tall.

Write an inequality for the rider's height h.

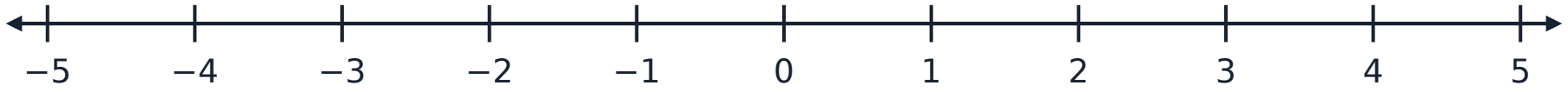
Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q1/5

Graph $x < 3$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q2/5

Which number is a solution of $x > -2$?

A. -2

B. -3

C. -5

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q3/5

True or false?

The inequality $x > 5$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q4/5

True or false?

On a number line, $x \leq 2$ is shown with an open circle at 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSA-REI.B.3

FORWARD 1

Use inequalities to meet constraints in a design problem • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A rectangular garden is 12 ft long. Its perimeter can be at most 50 ft.

Write and solve an inequality for the width w.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q1/5

A box with a 10 in by 10 in base must hold at least 600 in^3.

Write and solve an inequality for the height h.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q2/5

A poster 15 in wide must have an area of at least 300 in².

Which lengths L work?

A. L ≥ 285 inches

B. L ≥ 20 inches

C. L ≥ 4.5 inches

D. L ≤ 20 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q3/5

True or false?

A rectangle 8 cm long and w cm wide has an area of at most 40 cm² exactly when $0 < w \leq 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q4/5

True or false?

A square with side s cm has a perimeter less than 20 cm exactly when $s < 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSA-REI.B.3

FORWARD 2

Write a margin-of-error interval as a compound inequality • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: A margin-of-error interval is written as a compound inequality like those solved in the MAIN set.

A survey estimates that 54% of voters support a plan, with a margin of error of 3 percentage points.

Write the interval for the true proportion p as a compound inequality.

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.4 • Set 19 • FORWARD 2 • Q1/5 • NEAREST RELATED

A poll reports 62% with a margin of error of 4 percentage points.

Is 60% inside the interval? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions

HSS-IC.B.4 • Set 19 • FORWARD 2 • Q2/5 • NEAREST RELATED

The interval $0.40 \leq p \leq 0.48$ is an estimate plus or minus a margin of error.

What are the estimate and the margin of error?

A. 0.44 and 0.08

B. 0.40 and 0.08

C. 0.44 and 0.04

D. 0.48 and 0.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

An estimate of 70% with a margin of error of 5 percentage points gives $65\% \leq p \leq 75\%$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

If the interval is $0.30 \leq p \leq 0.36$, the margin of error is 0.06.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q5/5 • NEAREST RELATED

Create linear equations and inequalities in one variable and use them to solve problems

MAIN	Algebra I main question	**HSA-CED.A.1**
BACKWARD 1	Solve linear equations and inequalities	**HSA-REI.B.3**
BACKWARD 2	Write and solve one-step equations	**6.EE.B.7**
FORWARD 1	Write and solve equations from similar triangles	**HSG-SRT.B.5**
FORWARD 2	Write and solve an equation for an expected value — nearest relat.	**HSS-MD.A.2**

SET 20 • HSA-CED.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

A plumber charges a $65 call fee plus $48 per hour. A bill is $257.

Write an equation for the number of hours h, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q1/5

The sum of three consecutive integers is 141.

Write an equation and find the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q2/5

Maya has $180 and saves $15 each week. She wants at least $400.

Write and solve an inequality for the number of weeks w. How many whole weeks must she save?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q3/5

A rental costs $40 per day plus a one-time $25 fee.

Which inequality gives the number of days d you can rent for at most $265?

A. $40(d + 25) \leq 265$

B. $40d + 25 \geq 265$

C. $40d + 25 \leq 265$

D. $25d + 40 \leq 265$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q4/5

True or false?

Plan A costs 30 + 0.10m dollars and Plan B costs 0.25m dollars for m minutes. Plan A costs less when $m < 200$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q5/5

SET 20 • HSA-CED.A.1

BACKWARD 1

Solve linear equations and inequalities • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $7x - 4 = 3x + 20$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q1/5

Solve $2(5 - x) \geq 16$.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q2/5

Solve 0.25t + 3 = 7.

A. t = 1

B. t = 40

C. t = 2.5

D. t = 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q3/5

True or false?

$x = -3$ is a solution of $4 - x > 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q4/5

True or false?

The solution of $12 - 3x < 0$ is $x < 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSA-CED.A.1

BACKWARD 2

Write and solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve x + 17 = 42.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q1/5

Tickets cost $9 each. Sam spent $63.

Write an equation and find the number of tickets t.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q2/5

Solve 4.5 = 1.5y.

A. y = 6.75

B. y = 6

C. y = 3

D. y = 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q3/5

True or false?

If $n - 8 = 15$, then $n = 23$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q4/5

True or false?

If 6p = 3, then p = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSA-CED.A.1

FORWARD 1

Write and solve equations from similar triangles • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

A 6-ft person casts a 4-ft shadow at the same time a tree casts a 22-ft shadow.

Write a proportion and find the tree's height.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q1/5

Triangle ABC ~ triangle DEF with AB = 8, BC = 12, and DE = 10.

Find EF.

EF = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q2/5

Triangle PQR ~ triangle STU, with PQ = 5, ST = 15, QR = x + 2, and TU = 2x + 9.

What is x?

A. x = 1

B. x = 4.5

C. x = 3

D. x = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q3/5

True or false?

If two triangles are similar, their corresponding side lengths are proportional.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q4/5

True or false?

In similar triangles with a ratio of 2 : 3, a side of 10 corresponds to a side of 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSA-CED.A.1

FORWARD 2

Write and solve an equation for an expected value • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Writing and solving a linear equation from a situation is applied to an expected-value condition.

A game pays $10 with probability 0.2 and $0 otherwise.

What is the expected payout?

Expected payout: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q1/5 • NEAREST RELATED

A game pays x dollars with probability 0.25 and $0 otherwise.

Write and solve an equation for x so that the expected payout is $3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q2/5 • NEAREST RELATED

A random variable is 1 with probability 0.5, 2 with probability 0.3, and 4 with probability 0.2.

What is its expected value?

A. 7

B. 1.9

C. 2

D. 2.33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

The expected value of one roll of a fair number cube numbered 1 to 6 is 3.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A game costs $5 to play. It pays $20 with probability 0.2 and $0 otherwise. The expected net gain is $4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q5/5 • NEAREST RELATED

Rearrange formulas to highlight a quantity of interest

MAIN	Algebra I main question	**HSA-CED.A.4**
BACKWARD 1	Solve linear equations with letter coefficients	**HSA-REI.B.3**
BACKWARD 2	Evaluate formulas for given values	**6.EE.A.2.c**
FORWARD 1	Rearrange the arc-length formula $s = r\theta$	**HSG-C.B.5**
FORWARD 2	Solve exponential formulas for the exponent with logarithms	**HSF-LE.A.4**

SET 21 • HSA-CED.A.4

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

Solve $V = \pi r^2 h$ for h. ($r > 0$)

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q1/5

Solve $C = \frac{5}{9}(F - 32)$ for F.

F = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q2/5

Solve $A = P(1 + rt)$ for t. ($P \neq 0$, $r \neq 0$)

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q3/5

Ohm's law is $V = IR$. Solve for R. ($I \neq 0$)

A. $R = \frac{I}{V}$

B. $R = V - I$

C. $R = VI$

D. $R = \frac{V}{I}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q4/5

True or false?

Solving $y = mx + b$ for x ($m \neq 0$) gives $x = y \div m - b$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q5/5

SET 21 • HSA-CED.A.4

BACKWARD 1

Solve linear equations with letter coefficients • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $3x + k = 10$ for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q1/5

Solve $ax = bx + c$ for x, where $a \neq b$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q2/5

Solve $x \div m = n$ for x. ($m \neq 0$)

A. $x = m + n$

B. $x = mn$

C. $x = n \div m$

D. $x = m \div n$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q3/5

True or false?

If $2x - p = q$, then $x = (p + q) \div 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q4/5

True or false?

If $kx = 5$ and $k \neq 0$, then $x = 5k$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSA-CED.A.4

BACKWARD 2

Evaluate formulas for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Use A = lw to find A when l = 9 and w = 4.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q1/5

Use d = rt to find d when r = 55 and t = 3.

d = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q2/5

Use P = 2l + 2w to find P when l = 7 and w = 3.

A. 10

B. 21

C. 20

D. 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q3/5

True or false?

$V = s^3$ gives $V = 27$ when $s = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q4/5

True or false?

$A = \frac{1}{2}bh$ gives $A = 40$ when $b = 10$ and $h = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSA-CED.A.4

FORWARD 1

Rearrange the arc-length formula s = rθ • High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5

A circle has radius 6 cm. An arc has length 9 cm.

Use $s = r\theta$ to find the central angle θ in radians.

$\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q1/5

Solve $s = r\theta$ for r. Then find r when $s = 10\pi$ and $\theta = \pi \div 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q2/5

The area of a sector is $A = \frac{1}{2}r^2\theta$. Solve for θ. ($r > 0$)

A. $\theta = A - \frac{1}{2}r^2$

B. $\theta = 2Ar^2$

C. $\theta = A \div (2r^2)$

D. $\theta = 2A \div r^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q3/5

True or false?

An arc of length $2\pi r$ has a central angle of 2π radians.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q4/5

True or false?

If $s = r\theta$, then $\theta = r \div s$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSA-CED.A.4

FORWARD 2

Solve exponential formulas for the exponent with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $A = Pe^{rt}$ for t. (A, P, r > 0)

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q1/5

$500 grows to $1,000 with continuous interest at r = 0.05.

Find t to the nearest tenth of a year.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q2/5

Solve $2^x = y$ for x. ($y > 0$)

A. $x = y^2$

B. $x = \sqrt{y}$

C. $x = y \div 2$

D. $x = \log_2 y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q3/5

True or false?

If $10^k = 1{,}000$, then $k = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q4/5

True or false?

Solving $N = 5 \cdot 3^t$ for t gives $t = \log_3 N - 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q5**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q5/5

Complete the square to solve quadratic equations and derive the quadratic formula

MAIN	Algebra I main question	**HSA-REI.B.4.a**
BACKWARD 1	Complete the square to write a quadratic in vertex form	**HSA-SSE.B.3.b**
BACKWARD 2	Solve $x^2 = p$ and $x^3 = p$ with square and cube roots	**8.EE.A.2**
FORWARD 1	Complete the square to find the center and radius of a circle	**HSG-GPE.A.1**
FORWARD 2	Complete the square to write the equation of an ellipse	**HSG-GPE.A.3**

SET 22 • HSA-REI.B.4.a

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

Solve $x^2 + 6x - 7 = 0$ by completing the square.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q1/5

Rewrite $x^2 - 10x + 18 = 0$ in the form $(x - p)^2 = q$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q2/5

Rewrite $2x^2 + 12x + 5 = 0$ in the form $(x - p)^2 = q$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q3/5

Which number completes the square?

$x^2 + 8x + ___$

A. 8

B. 16

C. 4

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q4/5

True or false?

Completing the square on $ax^2 + bx + c = 0$ ($a \neq 0$) gives $(x + b \div (2a))^2 = (b^2 - 4ac) \div (4a^2)$, which leads to the quadratic formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q5/5

SET 22 • HSA-REI.B.4.a

BACKWARD 1

Complete the square to write a quadratic in vertex form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

Write $x^2 + 4x + 1$ in the form $(x - h)^2 + k$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q1/5

Write $x^2 - 6x + 11$ in the form $(x - h)^2 + k$, and give its minimum value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q2/5

Which expression is equal to $x^2 + 2x + 5$?

A. $(x + 2)^2 + 1$

B. $(x + 1)^2 + 5$

C. $(x + 1)^2 + 4$

D. $(x - 1)^2 + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q3/5

True or false?

$$(x - 4)^2 = x^2 - 8x + 16$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q4/5

True or false?

$x^2 + 10x = (x + 5)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSA-REI.B.4.a

BACKWARD 2

Solve $x^2 = p$ and $x^3 = p$ with square and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve $x^2 = 49$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q1/5

Solve $x^3 = -64$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q2/5

What are the solutions of $x^2 = 81$?

A. 9 and −9

B. 9 only

C. −9 only

D. 40.5 and −40.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{64} = 8$ and $\sqrt[3]{64} = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q4/5

True or false?

The only solution of $x^2 = 25$ is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSA-REI.B.4.a

FORWARD 1

Complete the square to find the center and radius of a circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Find the center and radius of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q1/5

Write an equation of the circle with center (2, −5) and radius 6 in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q2/5

What is the center of the circle $x^2 + 8x + y^2 = 9$?

A. $(-8, 0)$

B. $(-4, 0)$

C. $(4, 0)$

D. $(0, -4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q3/5

True or false?

The circle $(x - 2)^2 + y^2 = 16$ has radius 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q4/5

True or false?

The circle $x^2 + (y + 5)^2 = 9$ has center (0, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSA-REI.B.4.a

FORWARD 2

Complete the square to write the equation of an ellipse • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

Write $x^2 + 4y^2 - 6x - 7 = 0$ in standard form, and give the center of the ellipse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q1/5

Derive the equation of the set of points whose distances to the foci (3, 0) and (−3, 0) add to 10.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q2/5

The ellipse $(x - 1)^2 \div 25 + (y + 3)^2 \div 9 = 1$ has a horizontal major axis.

How long is the major axis?

A. 6

B. 5

C. 25

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F2-Q3**

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q3/5

True or false?

Completing the square on $x^2 - 8x$ gives $(x - 4)^2 - 16$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q4/5

True or false?

The ellipse $x^2 \div 4 + y^2 \div 9 = 1$ is wider than it is tall.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q5/5

Derive the quadratic formula by completing the square

MAIN	Algebra I main question	**HSA-REI.B.4.a**
BACKWARD 1	Complete the square in expressions	**HSA-SSE.B.3.b**
BACKWARD 2	Solve literal equations for a variable	**HSA-CED.A.4**
FORWARD 1	Use the quadratic formula when the discriminant is negative	**HSN-CN.C.7**
FORWARD 2	Count the complex solutions of quadratic equations	**HSN-CN.C.9**

SET 71 • HSA-REI.B.4.a

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

Derive the quadratic formula. Start from $ax^2 + bx + c = 0$ with $a \neq 0$, and complete the square to solve for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q1/5

The equation $x^2 + (b \div a)x = -c \div a$ comes from $ax^2 + bx + c = 0$. What must be added to both sides to complete the square, and what does the left side become?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q2/5

After completing the square,
$(x + b \div (2a))^2 = (b^2 - 4ac) \div (4a^2)$. Explain how taking square roots of both sides gives the ± and the denominator 2a.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q3/5

In the derivation of the quadratic formula, why must $a \neq 0$?

A. Because the discriminant must be positive.

B. Because b must be positive.

C. Because c must be 0.

D. The first step divides by a, and if $a = 0$ the equation is not quadratic.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q4/5

True or false?

The derivation shows that when $b^2 - 4ac < 0$, the equation has no real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q5/5

SET 71 • HSA-REI.B.4.a

BACKWARD 1

Complete the square in expressions • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

Complete the square.

$x^2 + 12x + 7$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q1/5

Complete the square.

$x^2 - 5x$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q2/5

What number completes the square for $x^2 + \frac{2}{3}x$?

A. $\frac{1}{9}$

B. $\frac{2}{3}$

C. $\frac{4}{9}$

D. $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q3/5

True or false?

$$x^2 + bx + (b \div 2)^2 = (x + b \div 2)^2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q4/5

True or false?

To complete the square on $2x^2 + 8x$, add 16 to the expression.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q5/5

SET 71 • HSA-REI.B.4.a

BACKWARD 2

Solve literal equations for a variable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

Solve $ax + b = 0$ for x. ($a \neq 0$)

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q1/5

Solve $(x + p)^2 = q$ for x. ($q \geq 0$)

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q2/5

Solve $y = kx^2$ for x, where $k > 0$ and $y \geq 0$.

A. $x = \pm\sqrt{(ky)}$

B. $x = \sqrt{y} \div k$

C. $x = \pm\sqrt{(y \div k)}$

D. $x = y \div k$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q3/5

True or false?

Solving $A = \pi r^2$ for r (with $r > 0$) gives $r = \sqrt{(A \div \pi)}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q4/5

True or false?

Solving $2x - c = d$ for x gives $x = (d - c) \div 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q5/5

SET 71 • HSA-REI.B.4.a

FORWARD 1

Use the quadratic formula when the discriminant is negative • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7

Use the quadratic formula to solve $x^2 + 6x + 13 = 0$. Write the solutions as $a + bi$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q1/5

Solve $2x^2 - 2x + 5 = 0$. Write the solutions as $a + bi$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q2/5

What is the discriminant of $3x^2 + x + 2$?

A. 25

B. -23

C. -25

D. 23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F1-Q3**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q3/5

True or false?

When $b^2 - 4ac < 0$, the quadratic formula gives two complex solutions that are conjugates.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q4/5

True or false?

The solutions of $x^2 + 9 = 0$ are 3 and -3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q5/5

SET 71 • HSA-REI.B.4.a

FORWARD 2

Count the complex solutions of quadratic equations • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9

How many complex solutions, counted with multiplicity, does every quadratic equation have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q1/5

$x^2 - 6x + 9 = 0$ has one real solution. How is it counted by the Fundamental Theorem of Algebra?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q2/5

A quadratic equation with real coefficients has the solution 2 + i. What is its other solution?

A. $-2 - i$

B. $2 - i$

C. $-2 + i$

D. 2 + i again

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q3/5

True or false?

Every quadratic equation with real coefficients has two solutions in the complex numbers, counted with multiplicity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q4/5

True or false?

A quadratic equation can have three different complex solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q5/5

Solve quadratic equations by inspection, square roots, and factoring

MAIN	Algebra I main question	**HSA-REI.B.4.b**
BACKWARD 1	Factor quadratic expressions	**HSA-SSE.A.2**
BACKWARD 2	Solve the linear equations that come from each factor	**HSA-REI.B.3**
FORWARD 1	Find the zeros of a polynomial from its factors	**HSA-APR.B.3**
FORWARD 2	Use the Fundamental Theorem of Algebra to count complex zeros	**HSN-CN.C.9**

SET 23 • HSA-REI.B.4.b

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve by factoring.

$x^2 - 5x - 24 = 0$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q1/5

Solve $3(x - 2)^2 = 75$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q2/5

Solve $2x^2 = 7x$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q3/5

Solve $6x^2 + x - 2 = 0$.

A. $x = 2$ or $x = -1$

B. $x = -\frac{2}{3}$ or $x = \frac{1}{2}$

C. $x = \frac{2}{3}$ or $x = -\frac{1}{2}$

D. $x = -2$ or $x = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q4/5

True or false?

A student divides both sides of $x^2 = 4x$ by x and gets $x = 4$. This finds every solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q5/5

SET 23 • HSA-REI.B.4.b

BACKWARD 1

Factor quadratic expressions • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor completely.

$x^2 + 2x - 15$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q1/5

Factor completely.

$4x^2 - 9$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q2/5

Factor $x^2 - 9x + 20$.

A. $(x - 4)(x + 5)$

B. $(x - 2)(x - 10)$

C. $(x - 4)(x - 5)$

D. $(x + 4)(x + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q3/5

True or false?

$3x^2 - 12x = 3x(x - 4)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q4/5

True or false?

$x^2 + 9 = (x + 3)(x - 3)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSA-REI.B.4.b

BACKWARD 2

Solve the linear equations that come from each factor • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $2x + 7 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q1/5

Solve $3x - 4 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q2/5

Solve $5 - 2x = 0$.

A. $x = 3$

B. $x = -2.5$

C. $x = 2.5$

D. $x = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q3/5

True or false?

If $x - 6 = 0$, then $x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q4/5

True or false?

If $4x = 0$, then $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q5/5